

A Dynamic Linear Model Approach to Time-Varying Cointegrated Systems

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*Fala-me, Musa, do homem astuto que tanto vagueou
depois que de Troia destruiu a cidadela sagrada.*

Agradecimentos

Agradeço ao meu orientador, Pedro Morettin, pelo tempo e pela paciência empregados na condução desta dissertação. Foi um grande prazer trabalhar com o senhor. Também sou grato ao IME-USP e ao seu corpo docente, ao CNPq e principalmente à minha família: Mauro, Patrícia e Laura.

Resumo

Fernando Medalha Rosa. **Uma Abordagem de Modelos Lineares Dinâmicos para Sistemas Cointegrados com Variação Temporal**. Dissertação (Mestrado). Instituto de Matemática e Estatística, Universidade de São Paulo, São Paulo, 2026.

Esta dissertação propõe uma nova formulação de Modelos Lineares Dinâmicos (DLM) para sistemas cointegrados com parâmetros variantes no tempo, baseada em uma representação de espaço de estados com recursões analíticas. Especificamente, buscamos construir um Modelo de Correção de Erros Vetorial (VECM) cujos parâmetros variam ao longo do tempo através da equação de estado. Contudo, uma aplicação direta do Filtro de Kalman e do Suavizador de Kalman é inviável devido ao posto reduzido do VECM, o que introduz a não linearidade em alguns parâmetros e problemas de identificação global. Para superar esses desafios, propomos um arcabouço DLM-VECM, o qual condiciona a matriz de ajuste a um valor fixo e impõe restrições à equação de estado para os parâmetros relacionados à matriz de cointegração. O resultado é uma representação linear em um sistema de espaço de estados que preserva a estrutura de posto reduzido, além de permitir que a evolução do espaço de cointegração seja identificável e apresente propriedades desejáveis.

Duas especificações são obtidas dentro do arcabouço DLM-VECM. A primeira formulação permite variação temporal apenas na matriz de cointegração, enquanto a segunda estende essa variação aos demais parâmetros de regressão do VECM. Adicionalmente, é desenvolvido um algoritmo *Expectation Maximization (EM)* para cada formulação. Por fim, aplicamos o arcabouço DLM-VECM para investigar relações de cointegração com variação temporal entre os preços de gasolina, diesel e etanol no mercado brasileiro varejista de combustíveis.

Palavras-chave: Modelos Lineares Dinâmicos. Parâmetros Variantes no Tempo. Cointegração. Modelo de Correção de Erros Vetorial. Filtro de Kalman. Suavizador de Kalman. *Expectation Maximization*. Arcabouço DLM-VECM. DLM-VECM Restrito. DLM-VECM Geral.

Abstract

Fernando Medalha Rosa. **A Dynamic Linear Model Approach to Time-Varying Cointegrated Systems**. Thesis (Master's). Institute of Mathematics and Statistics, University of São Paulo, São Paulo, 2026.

This thesis proposes a new Dynamic Linear Model (DLM) formulation for time-varying cointegrated systems based on analytical state space recursions. Specifically, we aim to design a Vector Error Correction Model (VECM) whose parameters vary over time through the state equation. However, a direct application of standard Kalman filtering and smoothing recursions is infeasible due to the reduced-rank structure of the VECM, which introduces parameter nonlinearity and global identification issues. To overcome these challenges, we propose a DLM-VECM framework that conditions on a fixed adjustment matrix and restricts the state equation for the cointegration-related parameters. The result is a linear state space representation that preserves the reduced-rank structure and allows the evolution of the cointegration space to be identified and to exhibit sensible properties.

Two specifications are obtained within the DLM-VECM framework. The first formulation allows time variation only in the cointegration matrix, while the second extends this time variation to the remaining VECM regression parameters. Additionally, an Expectation Maximization (EM) procedure is developed for each model. To conclude, we apply the DLM-VECM framework to investigate time-varying cointegration relationships among gasoline, diesel, and ethanol prices in the Brazilian retail fuel market.

Keywords: Dynamic Linear Models. Time-Varying Parameters. Cointegration. Vector Error Correction Model. Kalman Filter. Kalman Smoother. Expectation Maximization. DLM-VECM framework. Restricted DLM-VECM. General DLM-VECM.

List of Abbreviations

DLM	Dynamic Linear Model
VECM	Vector Error Correction Model
EM	Expectation Maximization
VAR	Vector Autorregression
TVP	Time-Varying Parameter
DMA	Dynamic Model Averaging
DFM	Dynamic Factor Model
KDE	Kernel Density Estimator
MSFE	Mean Squared Forecast Error

List of Symbols

B_t	Unrestricted Cointegration Matrix
b_t	Vectorized Unrestricted Cointegration Matrix
β_t	Semi-orthogonal Cointegration Matrix
$\boldsymbol{p}_t$	Cointegration Space
$\mathcal{N}$	Normal distribution
i.i.d.	Independent and Identically Distributed
p	Probability Density Function
L	Likelihood

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Chapter 1

Introduction

Time-varying cointegrated systems have received little attention in the time series literature. The existing methods were mainly developed through basis expansions or simulation within a Bayesian framework. For example, the related research on basis functions includes splines, Fourier series [PARK and HAHN, 1999], Chebyshev polynomials [BIERENS and MARTINS, 2010], and Wavelets [FONSECA *et al.*, 2019]. With respect to simulation-based approaches, the main contributions are the Bayesian state space formulation of KOOP, LEON-GONZALEZ, *et al.*, 2011 and the regime-switching framework of JOCHMANN and KOOP, 2015.

Static cointegrated systems, in contrast, are well established in the literature. Since the contributions of JOHANSEN, 1988; JOHANSEN, 1991, direct modelling of the error structure through Vector Error Correction Models (VECMs) has emerged as an alternative to Vector Autoregressions (VARs) in levels when the underlying system presents common stochastic trends. In that sense, estimates that consider the error structure are believed to behave better than regression estimates that do not [JOHANSEN, 1988].

Additionally, it may be desirable to allow for time-varying parameters (TVP) in multivariate processes. In this context, state space modelling through Dynamic Linear Models (DLMs) provides an attractive solution. Such developments for VARs can be found in the Bayesian formulations of COGLEY and SARGENT, 2001; COGLEY and SARGENT, 2005, PRIMICERI, 2005, KOOP, KOROBILO, *et al.*, 2010; KOOP and KOROBILO, 2013. For VECMs, the only existing approach is the Bayesian specification of KOOP, LEON-GONZALEZ, *et al.*, 2011.

While Bayesian VAR formulations allow for DLM analytical computations through a suitable choice of priors [e.g., the conjugate Minnesota prior of LITTERMAN, 1986], the same does not hold for VECMs, where closed-form solutions for filtering and smoothing are not available. This difficulty arises from the reduced-rank structure of the VECM, which introduces two nontrivial problems: parameter nonlinearity and the lack of global identification.

With respect to parameter nonlinearity, this feature prohibits the direct application of analytical Kalman filtering and smoothing. Our proposed solution to this first problem is to condition the observation equation on a fixed adjustment matrix. As a consequence,

the state space representation becomes linear in the parameters, and the Kalman filtering and smoothing formulas become readily applicable.

The other source of concern is related to the lack of global identification of the cointegration matrix. In the absence of identifying restrictions, only the cointegration space is identified, which motivates characterizing time variation at the cointegration space rather than in the unnormalized cointegration vectors. A naive strategy would then be to incorporate a constrained cointegration matrix into the DLM. However, this is not possible, since linear state space models do not accommodate the usual cointegration identifying restrictions (KOOP, LEON-GONZALEZ, *et al.*, 2011).

This raises a cumbersome problem: how can time variation of the cointegration space be modelled within a DLM representation? One solution to this question is provided by KOOP, LEON-GONZALEZ, *et al.*, 2011. They demonstrate that a DLM formulation for the unnormalized cointegration vectors, combined with simple restrictions on the parameters of the state equation, leads to an appealing evolution of the cointegration space over time. Specifically, it implies that the evolution of the cointegration space at time t is gradual and centered over the cointegration space at time $t - 1$. Therefore, their proposed dynamics for the cointegration space closely resembles that of a DLM-VAR with a random-walk in the state equation.

Given the aforementioned nonlinearity and identification-related concerns, this thesis proposes a feasible Dynamic Linear Model (DLM) formulation that addresses these issues and enables analytical state space modelling for Vector Error Correction Models. The result is a linear state space representation that preserves the reduced-rank structure of the VECM and accounts for the evolution of the underlying cointegration space in a sensible manner. The overall approach is referred to as the DLM-VECM framework.

Accordingly, introducing time variation through the state equation becomes straightforward under the proposed treatment. However, it remains to specify which parameters may vary stochastically over time. In this context, we define two classes of models within the DLM-VECM framework. The first formulation enables time variation only in the cointegration matrix, while the second also extends it to the remaining VECM regression parameters. These models are named as the Restricted DLM-VECM and the General DLM-VECM, respectively.

In order to proceed with estimation, we develop an Expectation Maximization (EM) algorithm for each version of the DLM-VECM. In addition to the traditional DLM parameters,¹ the two proposed EM algorithms also update the adjustment matrix. The key difference between them is the following: while the Restricted DLM-VECM EM algorithm further updates the remaining VECM parameters,² the General DLM-VECM EM algorithm treats these parameters as time-varying states. Furthermore, we also complement these EM procedures with a constrained M-step in order to achieve full identification of the system.

Having established the theory behind the analytical DLM-VECM framework, we proceed to answer the following questions: Do our models perform well in empirical

¹ The state transition matrix, covariance matrices for both the state and observation equations, and initial states along with their covariance matrix.

² That is, the parameters other than the adjustment matrix and the cointegration matrix.

applications? What are their limitations? And how could we overcome these limitations in order to improve performance?

To answer these questions, we conduct an empirical application of the DLM-VECM framework using weekly data on gasoline, diesel, and ethanol prices in the Brazilian retail fuel market from May 2004 to November 2025. We begin by estimating both the Restricted DLM-VECM and the General DLM-VECM. In these settings, the Restricted DLM-VECM displays good predictive performance and a meaningful evolution of the cointegration space over time, while the General DLM-VECM becomes too flexible and leads to overfitting.

From this application, we infer that the excessive flexibility of the General DLM-VECM may lead to its failure. Accordingly, overcoming this limitation would require setting additional restrictions on the model. Nevertheless, a careful investigation of these restrictions would blur the scope of this thesis and is left for future research.

This thesis is structured as follows. Chapter 2 develops the DLM-VECM framework. Chapter 3 introduces the Restricted DLM-VECM model and its corresponding EM algorithm. Chapter 4 presents the General DLM-VECM model, along with its associated EM algorithm. Chapter 5 applies the DLM-VECM framework to the fuel price dataset. Chapter 6 concludes.

Chapter 2

The DLM-VECM Framework

This chapter is dedicated to introducing a new framework for time-varying cointegration. To this end, we address the challenges that arise when incorporating a VECM into a DLM specification, thereby proposing an appropriate set of solutions. To simplify our exposition, we consider a Basic DLM-VECM formulation that retains only the adjustment and cointegration matrices. Extensions that account for the remaining regression parameters are straightforward to implement and are presented in Chapters 3 and 4, along with their corresponding EM algorithms. The overall approach developed in this chapter, along with its extensions and associated EM algorithms, constitutes what we refer to as the DLM-VECM framework.

2.1 Introduction

Consider the following VECM formulation with time-varying parameters:

$$\Delta y_t = c_t + A_t B_t^\top y_{t-1} + \sum_{i=1}^{p-1} \Gamma_{i,t} \Delta y_{t-i} + \Phi_t d_t + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, H), \quad (2.1)$$

where A_t and B_t are full rank matrices of dimension $n \times r$. The quantities n , r and $p - 1$ represent the number of variables, the number of cointegration relationships and the number of lags, respectively. Additionally, d_t denotes q deterministic components.

A suitable way of addressing time variation in the above parameters is to extend the so-called observation equation (2.1) with a state equation that specifies the parameters as states and allows them to evolve as an autoregressive process. However, the term $A_t B_t^\top$ precludes the application of standard dynamic linear modelling due to nonlinearity and identification issues.

In this context, KOOP, LEÓN-GONZÁLEZ, *et al.*, 2011 develop a feasible Bayesian state space formulation for a TVP-VECM. Their work proposes an efficient algorithm for sampling the posterior distribution of a TVP-VECM through the combination of the simulation smoother of DURBIN and KOOPMAN, 2002 and the efficient sampler of KOOP, LEÓN-GONZÁLEZ, *et al.*, 2009. To tackle the nonlinear VECM structure, they formulate two

complementary DLMs: one for the parameters associated with the cointegration matrix B_t , and another for the remaining parameters of the VECM ($c_t, A_t, \Gamma_{1,t}, \dots, \Gamma_{p-1,t}, \Phi_t$). Then, linearity is achieved by conditioning each model on the other, which results in two DLMs with stochastic measurement matrices.¹

Within a Bayesian framework, it is feasible to work with stochastic measurement matrices; however, this does not extend to traditional Kalman filtering and smoothing recursions. This is because stochastic measurement matrices violate the assumptions for analytical state space modelling, since the linear-normal structure of the DLM is no longer preserved. As a result, the standard formulas presented in Appendix B are not valid and the EM algorithm for DLMs does not converge monotonically.²

Consequently, it is not possible to simultaneously model time variation in both the adjustment matrix and the cointegration matrix within an analytical state space framework. Therefore, one of the matrices in the product $A_t B_t$ must be conditioned on in order to achieve linearity. Since our primary interest lies in modeling time-varying cointegration, we condition on the adjustment matrix and set it as a fixed parameter of our model.

In addition to the nonlinearity problem, it remains to propose a solution to the lack of global identification in TVP-VECM models. Since the unrestricted cointegration vectors B_t are not identifiable, our main interest lies in modelling time variation in the cointegration space, which is denoted by $\mathbf{p}_t = \text{sp}(B_t)$.³ In this context, our proposed DLM-VECM framework incorporates the developments of KOOP, LEON-GONZALEZ, *et al.*, 2011, which lead to a time evolution of $\mathbf{p}_t$ that is gradual and centered around the previous cointegration space $\mathbf{p}_{t-1}$.⁴ Observe that these properties resemble the traditional random-walk evolution of parameters in a linear state space model.

Complementarily, we also consider imposing exact restrictions on the adjustment matrix to achieve full identification of the DLM-VECM system. This procedure eliminates the remaining indeterminacies in the product $A_t B_t$, resulting in a unique representation of the system. As we demonstrate further, these additional restrictions can improve the numerical properties of the proposed EM algorithms.

As previously outlined, a Bayesian representation for TVP-VECMs has already been developed in the literature. However, in some cases, it may be preferable to work with an analytical alternative. For example, simulation-based approaches can present computationally demanding challenges, especially in high-dimensional settings. Another example relates to the difficulty of extending the Dynamic Model Averaging (DMA) framework [RAFTERY *et al.*, 2010; KOOP and KOROBILIS, 2012; KOOP and KOROBILIS, 2013; KOOP and KOROBILIS, 2014] to cointegrated systems.

Accordingly, we intend to fill the analytical gap in the time-varying cointegration

¹ Consider the term $A_t B_t^\top y_{t-1}$ in (2.1). Conditioning on B_t results in a model linear in A_t , since $X_t = B_t^\top y_{t-1}$. Accordingly, conditioning on A_t yields linearity in B_t , since the system may be equivalently expressed with $\text{vec}(A_t B_t^\top y_{t-1}) = (y_{t-1}^\top \otimes A_t) \text{vec}(B_t^\top) = \tilde{X}_t b_t$. Notice that both cases display stochastic measurement matrices (X_t and $\tilde{X}_t$).

² SHUMWAY and STOFFER, 2006, chapter 6.

³ The operator sp denotes the space spanned by the columns of B_t .

⁴ See Propositions 1–4 of KOOP, LEON-GONZALEZ, *et al.*, 2011 in Appendix C.

literature by developing a DLM-VECM framework that allows for Kalman filtering and smoothing recursions in cointegrated systems. The remainder of this chapter presents the basic skeleton of the DLM-VECM model, focusing exclusively on the adjustment and cointegration matrices. It also develops the corresponding EM algorithm for the basic specification and derives the case in which additional restrictions are imposed on the adjustment matrix to achieve full identification. Further extensions are considered in Chapters 3 and 4.

2.2 The Basic DLM-VECM model

To simplify our exposition, we set the parameters of equation (2.1) associated with the intercept, lag matrices and deterministic terms to zero, leading to

$$\Delta y_t = A_t B_t^\top y_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, H), \quad (2.2)$$

If this simplified equation is conditioned on the adjustment matrix A_t , then it becomes linear in the parameters. Additionally, we opt to specify the adjustment matrix as fixed over time. Therefore, applying the vec operator and dropping the subscript t from A_t results in

$$\begin{aligned} \Delta y_t &= \text{vec}(A B_t^\top y_{t-1}) + \varepsilon_t = (y_{t-1}^\top \otimes A) \text{vec}(B_t^\top) + \varepsilon_t \\ &= \tilde{X}_t b_t + \varepsilon_t \end{aligned} \quad (2.3)$$

where $\tilde{X}_t = (y_{t-1}^\top \otimes A)$, $b_t = \text{vec}(B_t^\top)$ and $\varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, H)$. With the objective of introducing time-varying parameters, system (2.3) can be transformed into the following Dynamic Linear Model:

$$\Delta y_t = \tilde{X}_t b_t + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, H), \quad (2.4)$$

$$b_t = \rho b_{t-1} + \eta_t, \quad \eta_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_{nr}), \quad (2.5)$$

where $\rho \in [0, 1)$ is an overall autoregressive coefficient and I_{nr} stands for an isotropic state covariance matrix. As demonstrated in [KOOP, LEON-GONZALEZ, et al., 2011](#), this choice of parameters in the state equation implies that the conditional distribution of the time-varying cointegration space, denoted by $\mathbf{p}_t \mid \mathbf{p}_{t-1}$, is centered around $\mathbf{p}_{t-1}$ and has finite concentration.⁵

Given the observation equation in (2.4) and the state equation in (2.5), it remains to specify the initial distribution of the state vector b_t . We assume

$$b_0 \sim \mathcal{N}(0, (1 - \rho^2)^{-1} I_{nr}), \quad (2.6)$$

which corresponds to the unconditional distribution of a stationary AR(1) process. The initial state, denoted by $b_{0|0}$, can either be sampled from this distribution or set to suit-

⁵ Propositions 1 and 2 in the cited paper state that the mode and expected value of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ under the proposed parameters are both given by $\mathbf{p}_{t-1}$. Moreover, Propositions 3 and 4 discuss about the concentration properties of the cointegration space, thereby establishing that the evolution of $\mathbf{p}_{t-1}$ to $\mathbf{p}_t$ has finite variance if $\rho \in [0, 1)$. These results are also displayed in Appendix C.

able values in order to initialize the state space recursions. Accordingly, the initial state covariance is given by $\Sigma_{0|0}^{(b)} = (1 - \rho^2)^{-1} I_{nr}$.

2.2.1 More on the State Equation Parameters

Before proceeding, it is important to discuss some further implications that arise from the state equation parameters $\rho \in [0, 1)$ and I_{nr} . Starting with ρ , the case $\rho = 1$ is ruled out because it implies that the expected concentration of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ is infinite, leading to a degenerate distribution that shrinks the variation of the cointegration space to zero in the limit. To verify this claim, see item 2 of Proposition 4 in Appendix C.

Moreover, it is ρ that controls the variability of the state equation, as the state error covariance matrix is the identity I_{nr} . Appendix D showcases a simulation exercise that illustrates the effect of ρ on the variability of the cointegration space. After examining this Appendix section, it is easy to conclude that, for most economic systems, ρ must assume values that are very close to one. As a result, we restrict the value of ρ to lie between 0.999 and 0.9999999 in the M-step of the EM algorithms proposed in Chapters 2, 3 and 4.

In addition, the isotropic state covariance matrix I_{nr} results in desirable identification implications. Without any identification restrictions, the product $AB_t^\top$ is invariant for any invertible matrix $R_{r \times r}$, since

$$AB_t^\top = (AR^{-1})(RB_t^\top),$$

which introduces r^2 additional degrees of freedom. In contrast, an isotropic state covariance matrix I_{nr} leads to a matrix product $AB_t^\top$ that is invariant only to orthogonal rotations. To verify this claim, define $\tilde{B}_t^\top = RB_t^\top$, $\tilde{b}_t = \text{vec}(\tilde{B}_t^\top) = (I_n \otimes R)b_t$ and $\tilde{\eta}_t = (I_n \otimes R)\eta_t$. Then, the state equation for $\tilde{b}_t$ is given by:

$$\begin{aligned} \tilde{b}_t &= (I_n \otimes R)(\rho b_{t-1} + \eta_t) \\ &= \rho(I_n \otimes R)b_{t-1} + (I_n \otimes R)\eta_t \\ &= \rho \tilde{b}_{t-1} + \tilde{\eta}_t \end{aligned}$$

Since $\eta_t \sim \mathcal{N}(0, I_{nr})$, we have⁶

$$\begin{aligned} \text{Var}(\tilde{\eta}_t) &= (I_n \otimes R)\text{Var}(\eta_t)(I_n \otimes R)^\top \\ &= (I_n \otimes R)I_{nr}(I_n \otimes R)^\top \\ &= (I_n \otimes R)(I_n \otimes I_r)(I_n \otimes R)^\top \\ &= (I_n I_n I_n) \otimes (R I_r R^\top) \\ &= I_n \otimes (R R^\top). \end{aligned}$$

Accordingly, $\text{Var}(\tilde{\eta}_t) = I_{nr}$ only if $RR^\top = I_r$. Therefore, an isotropic state covariance matrix I_{nr} implies that R must be orthogonal, which reduces the number of unidentified degrees

⁶ The fourth equality follows from the mixed-product property of the Kronecker product, in which $(A \otimes B)(C \otimes D) = (AC \otimes BD)$.

of freedom from r^2 to $r(r-1)/2$.⁷ Notice that this magnitude of reduction in degrees of freedom is equivalent to that implied by a semi-orthogonal normalization on B_t .⁸

Furthermore, this reduction in degrees of freedom has important optimization implications for the DLM-VECM EM algorithms. In short, it reduces overparameterization and removes scale degeneracy, thereby leading to an optimization procedure that is less sensitive to poor initialization and less subject to numerical problems. Moreover, as previously outlined,⁹ the isotropic state covariance matrix I_{nr} also contributes to an evolution of the cointegration space over time that displays desirable properties.

2.2.2 Kalman Filtering and Smoothing in the Basic DLM-VECM

Given the Basic DLM-VECM formulation in equations (2.4)–(2.6), the standard Kalman filtering and smoothing analytical recursions are readily applicable. Appendix B contains the corresponding formulas, which require substituting y_t , M_t , θ_t , Φ , P , and Q with Δy_t , $\tilde{X}_t$, b_t , ρ , I_{nr} , and I_{nr} , respectively. Notice that these recursions are only feasible if we condition on the respective set of parameters $\Theta_{bas} = \{A, H, \rho\}$ and prespecify the cointegration rank r .

Algorithm 1 in Appendix A displays the filtering procedure for the Basic DLM-VECM, which outputs the sequences $\{b_{t|t-1}, \Sigma_{t|t-1}^{(b)}, b_{t|t}, \Sigma_{t|t}^{(b)}\}_{t=0}^T$ and $\{\tilde{X}_t\}_{t=1}^T$. Algorithm 2, in turn, summarizes the smoothing process, yielding $\{b_{t|T}, \Sigma_{t|T}^{(b)}\}_{t=0}^T$ and $\{\Sigma_{t,t-1|T}^{(b)}\}_{t=1}^T$.

2.3 Basic DLM-VECM EM algorithm

The Expectation Maximization (EM) algorithm is used to compute maximum likelihood estimates for models with latent variables. In the present case of the Basic DLM-VECM, we are interested in iteratively updating the parameter set $\Theta_{bas} = \{\rho, H, A\}$. After initialization, the EM algorithm alternates between the E-step, which computes the smoothed states given the current value of Θ_{bas} , and the M-step, which updates Θ_{bas} given the smoothed states. With respect to the general convergence results of the EM algorithm, we refer to Wu, 1983. Accordingly, the proposed EM algorithm generates a non-decreasing sequence of observed likelihood values.

We begin by presenting the corresponding Unconstrained EM algorithm, which does not impose any restrictions on the elements of Θ_{bas} . Next, we present the associated Constrained EM algorithm, which imposes additional restrictions on the adjustment matrix A to achieve full identification of the system.

⁷ Since the restriction $RR^\top = I_r$ imposes $r(r+1)/2$ symmetry constraints defining orthogonality, the number of free parameters is now $r^2 - r(r+1)/2 = r(r-1)/2$.

⁸ $B_t^\top B_t = I_r$

⁹ Appendix C

2.3.1 Unconstrained EM algorithm

Our EM presentation begins by defining the complete-data joint density of the Basic DLM-VECM¹⁰:

$$p_{\Theta_{bas}}(b_{0:T}, \Delta y_{1:T}) = p_{b_{0|0}, \Sigma_{0|0}^{(b)}}(b_0) \prod_{t=1}^T p_{\rho}(b_t | b_{t-1}) \prod_{t=1}^T p_{A,H}(\Delta y_t | b_t), \quad (2.7)$$

where the first term is ruled out, since the sampled $b_{0|0}$ and $\Sigma_{0|0}^{(b)}$ are assumed fixed throughout the iterations of the EM algorithm. Notice that the above factorization follows from the conditional independence property of first-order Markov systems. From the Gaussianity assumptions in (2.4), (2.5) and (2.6), the remaining distributions can be denoted as $b_t | b_{t-1} \sim \mathcal{N}(\rho b_{t-1}, I_{nr})$ and $\Delta y_t | b_t \sim \mathcal{N}(\tilde{X}_t b_t, H)$. Ignoring constants, the complete-data log-likelihood of equation (2.7) is then formulated as

$$\begin{aligned} -2 \ln L(\Theta_{bas}) &= \sum_{t=1}^T (b_t - \rho b_{t-1})^\top (b_t - \rho b_{t-1}) + T \ln |H| \\ &\quad + \sum_{t=1}^T (\Delta y_t - \tilde{X}_t b_t)^\top H^{-1} (\Delta y_t - \tilde{X}_t b_t). \end{aligned} \quad (2.8)$$

E-step

If the states $\{b_t\}_{t=0}^T$ were available, the Maximum Likelihood Estimates of Θ_{bas} could be directly obtained from minimizing (2.8). However, since only $\{y_t\}_{t=0}^T$ is observed, we must work with the expected conditional log-likelihood, given by:

$$Q(\Theta_{bas} | \Theta_{bas}^{\text{old}}) = \mathbb{E}[-2 \ln L(\Theta_{bas}) | y_{0:T}, \Theta_{bas}^{\text{old}}], \quad (2.9)$$

where $\Theta_{bas}^{\text{old}}$ is associated with the previous iteration of the EM procedure. Calculation of the expectations in (2.9) stands for the E-step of the Basic DLM-VECM Unconstrained EM algorithm:

$$\begin{aligned} Q(\Theta | \Theta_{bas}^{\text{old}}) &= \text{tr} \left\{ \sum_{t=1}^T \mathbb{E}[(b_t - \rho b_{t-1})(b_t - \rho b_{t-1})^\top | y_{0:T}, \Theta_{bas}^{\text{old}}] \right\} + T \ln |H| \\ &\quad + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \mathbb{E}[(\Delta y_t - \tilde{X}_t b_t)(\Delta y_t - \tilde{X}_t b_t)^\top | y_{0:T}, \Theta_{bas}^{\text{old}}] \right\}. \end{aligned} \quad (2.10)$$

The expectations in the first and second line of (2.10) are denoted by the following expressions:

$$\begin{aligned} \mathbb{E}(b_t | y_{0:T}, \Theta_{bas}^{\text{old}}) &= b_{t|T}, \\ \mathbb{E}(b_s b_t' | y_{0:T}, \Theta_{bas}^{\text{old}}) &= b_{s|T} (b_{t|T})^\top + \Sigma_{s,t|T}^{(b)}, \end{aligned}$$

¹⁰ Equations (2.4), (2.5) and (2.6).

and their evaluation is straightforward. Accordingly, if we define the moment matrices

$$\begin{aligned} S_{11}^{(b)} &= \sum_{t=1}^T \left(b_{t|T} (b_{t|T})^\top + \Sigma_{t|T}^{(b)} \right), \\ S_{10}^{(b)} &= \sum_{t=1}^T \left(b_{t|T} (b_{t-1|T})^\top + \Sigma_{t,t-1|T}^{(b)} \right), \\ S_{00}^{(b)} &= \sum_{t=1}^T \left(b_{t-1|T} (b_{t-1|T})^\top + \Sigma_{t-1|T}^{(b)} \right), \end{aligned}$$

equation (2.10) may be expressed as:

$$\begin{aligned} Q(\Theta_{bas} | \Theta_{bas}^{\text{old}}) &= \text{tr} \left[S_{11}^{(b)} - \rho S_{10}^{(b)} - \rho S_{10}^{(b)\top} + \rho^2 S_{00}^{(b)} \right] + T \ln |H| \\ &+ \text{tr} \left\{ H^{-1} \sum_{t=1}^T \left[\Delta y_t \Delta y_t^\top - 2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top + \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top \right] \right\}. \end{aligned} \quad (2.11)$$

M-step

Unrestricted maximization of (2.11) with respect to Θ_{bas} corresponds to the M-step of the Basic DLM-VECM Unconstrained EM algorithm, which updates the parameters ρ , H and A without further constraints. The corresponding update formulas for ρ and H are given by

$$\rho_{ur}^{\text{new}} = \left(\text{tr}(S_{10}^{(b)}) \right) \left(\text{tr}(S_{00}^{(b)}) \right)^{-1}, \quad (2.12)$$

and

$$H_{ur}^{\text{new}} = \frac{1}{T} \sum_{t=1}^T \left(\Delta y_t \Delta y_t^\top - 2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top + \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top \right). \quad (2.13)$$

In order to perform the last maximization with respect to A , we first apply the identity

$$\tilde{X}_t = (y_{t-1}^\top \otimes A) = A(y_{t-1}^\top \otimes I_r).$$

Then we rewrite the terms of equation (2.11) that involve $\tilde{X}_t$ as:

$$\begin{aligned} -2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top &= -2 \Delta y_t b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top A^\top \\ \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top &= A(y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top A^\top \end{aligned}$$

To simplify our computations, we define

$$\tilde{S}_t = (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top, \quad (2.14)$$

$$\tilde{R}_t = \Delta y_t b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top. \quad (2.15)$$

Then, the part of (2.11) that depends on A can be written as

$$\begin{aligned} Q(A \mid \Theta_{bas}^{\text{old}}) &= \sum_{t=1}^T \text{tr} \left[H^{-1} (A \tilde{S}_t A^\top - 2A \tilde{R}_t^\top) \right] + \text{const}, \\ &= \text{tr} \left[H^{-1} (A \tilde{S} A^\top - 2A \tilde{R}^\top) \right] + \text{const}, \\ &= \text{tr} (A \tilde{S} A^\top H^{-1}) - \text{tr} (2A \tilde{R}^\top H^{-1}) + \text{const}, \end{aligned} \quad (2.16)$$

where $\tilde{S} = \sum_{t=1}^T (\tilde{S}_t)$ and $\tilde{R} = \sum_{t=1}^T (\tilde{R}_t)$. Therefore, the unrestricted maximization of (2.16) with respect to A yields:

$$\begin{aligned} A &= \tilde{R} \tilde{S}^{-1} \\ &= \left(\sum_{t=1}^T \Delta y_t b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top \right) \times \left(\sum_{t=1}^T (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top \right)^{-1}. \end{aligned} \quad (2.17)$$

2.3.2 Constrained EM algorithm

As previously outlined in Section 2.2.1, the expected conditional log-likelihood in equation (2.11) is invariant to any simultaneous linear transformation of the form $b_t \rightarrow (I_n \otimes R)b_t$ and $\tilde{X}_t \rightarrow \tilde{X}_t(I_n \otimes R)^{-1}$, in which R is an orthogonal $r \times r$ matrix. Therefore, the parameter sets $\Theta_{bas} = \{\rho, H, A\}$ and $\Theta_{bas,R} = \{\rho, H, AR^{-1}\}$ are observationally equivalent and lead to the same likelihood value with the corresponding linear transformation on the states.

This lack of identification and its implications for the likelihood find a parallel in the Dynamic Factor Model (DFM) literature. For instance, [BAÑBURA and MODUGNO, 2014](#) leverage [DEMPSTER *et al.*, 1977](#) to argue that the related DFM EM algorithm converges to a particular $\Theta_{bas,R}$ on the ridge of the likelihood function, rather than moving indefinitely between different points in the ridge.

Therefore, if the interest lies in forecasting and/or in estimating the space generated by the states, this lack of identifiability does not represent a problem, provided that the system is properly initialized. However, if the objective is to achieve direct identification of the decomposition $AB_t^\top$, $r(r-1)/2$ additional restrictions must be specified in order to remove the remaining orthogonal indeterminacies. In this case, a natural choice is to impose linear restrictions on the adjustment matrix A , which conveniently results in a closed-form solution for the corresponding M-step.

The simplest way to implement these $r(r-1)/2$ linear restrictions in the M-step for the adjustment matrix is to set zero restrictions on the upper-right block of the matrix A and to rewrite the E-step in terms of $a = \text{vec}(A)$. Consequently, the linear restrictions take the form

$$C_A a = 0, \quad (2.18)$$

where C_A is a $r(r-1)/2 \times nr$ matrix of zeros and ones that only extracts the elements of A associated with the zero restrictions.

E-step

To obtain the closed-form solutions that incorporate these restrictions, we first rewrite the expected conditional log-likelihood of equation (2.11) in terms of A , which yields:

$$\begin{aligned} Q(\Theta_{bas} \mid \Theta_{bas}^{\text{old}}) &= \text{tr} \left[S_{11}^{(b)} - \rho S_{10}^{(b)} - \rho S_{10}^{(b)\top} + \rho^2 S_{00}^{(b)} \right] + T \ln |H| \\ &\quad + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \left[\Delta y_t \Delta y_t^\top - 2\tilde{R}_t A^\top + A \tilde{S}_t A^\top \right] \right\}. \end{aligned} \quad (2.19)$$

Next, we express this conditional log-likelihood in terms of $a = \text{vec}(A)$. Using the recursions¹¹

$$\begin{aligned} \text{tr} (H^{-1} A \tilde{S} A^\top) &= \text{tr} (A^\top H^{-1} A \tilde{S}) = \text{vec}(A)^\top (\tilde{S}^\top \otimes H^{-1}) \text{vec}(A) \\ &= a^\top (\tilde{S}^\top \otimes H^{-1}) a, \\ 2 \text{tr} (H^{-1} \tilde{R} A^\top) &= 2 \text{tr} (A^\top H^{-1} \tilde{R}) = 2 \text{vec}(A)^\top \text{vec} (H^{-1} \tilde{R}) \\ &= 2a^\top \text{vec} (H^{-1} \tilde{R}), \end{aligned}$$

we may rewrite $Q(\Theta_{bas} \mid \Theta_{bas}^{\text{old}})$ as

$$\begin{aligned} Q(\Theta_{bas} \mid \Theta_{bas}^{\text{old}}) &= \text{tr} \left[S_{11}^{(b)} - \rho S_{10}^{(b)} - \rho S_{10}^{(b)\top} + \rho^2 S_{00}^{(b)} \right] + T \ln |H| \\ &\quad + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \Delta y_t \Delta y_t^\top \right\} - 2a^\top \text{vec} (H^{-1} \tilde{R}) + a^\top (\tilde{S}^\top \otimes H^{-1}) a. \end{aligned} \quad (2.20)$$

In this case, the E-step is given by the same expected conditional log-likelihood as in (2.11) or (2.16). The only difference is that it is rewritten in terms of a instead of $\tilde{X}_t$ or A .

M-step

The M-step now maximizes (2.20) with respect to $\{\rho, H, a\}$ subject to the linear restriction $C_A a = 0$. The objective function of the constrained optimization is given by the following Lagrangian:

$$\mathcal{L}(a, \lambda) = a^\top (\tilde{S}^\top \otimes H^{-1}) a - 2a^\top \text{vec} (H^{-1} \tilde{R}) + 2\lambda^\top C_A a, \quad (2.21)$$

where the first-order conditions can be written as

$$\frac{\partial \mathcal{L}(a, \lambda)}{\partial a} = 2(\tilde{S}^\top \otimes H^{-1}) a - 2 \text{vec} (H^{-1} \tilde{R}) + 2C_A^\top \lambda = 0, \quad (2.22)$$

$$\frac{\partial \mathcal{L}(a, \lambda)}{\partial \lambda} = 2C_A a = 0. \quad (2.23)$$

¹¹ For two matrices M and N with compatible dimensions, $\text{tr}(MN) = \text{vec}(M^\top)^\top \text{vec}(N)$. Additionally $\text{tr}(MN) = \text{tr}(NM)$ and $\text{tr}(M) = \text{tr}(M^\top)$.

Solving for a in (2.22) leads to

$$a = (\tilde{S}^\top \otimes H^{-1})^{-1} [\text{vec}(H^{-1}\tilde{R}) - C_A^\top \lambda]. \quad (2.24)$$

Now substituting (2.24) into the constraint $C_A a = 0$ results in

$$C_A (\tilde{S}^\top \otimes H^{-1})^{-1} \text{vec}(H^{-1}\tilde{R}) - C_A (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \lambda = 0, \quad (2.25)$$

which implies

$$\begin{aligned} \lambda &= \left[C_A (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \right]^{-1} C_A (\tilde{S}^\top \otimes H^{-1})^{-1} \text{vec}(H^{-1}\tilde{R}) \\ &= \left[C_A (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \right]^{-1} C_A a_{ur}, \end{aligned} \quad (2.26)$$

where

$$\begin{aligned} a_{ur} &= (\tilde{S}^\top \otimes H^{-1})^{-1} \text{vec}(H^{-1}\tilde{R}) = ((\tilde{S}^\top)^{-1} \otimes H) \text{vec}(H^{-1}\tilde{R}) \\ &= \text{vec}(H(H^{-1}\tilde{R})((\tilde{S}^\top)^{-1})^\top) = \text{vec}(\tilde{R}((\tilde{S}^\top)^{-1})^\top) = \text{vec}(\tilde{R}\tilde{S}^{-1}) \\ &= \text{vec}(A_{ur}) \end{aligned}$$

is the vectorized version of the unrestricted adjustment matrix. Finally, substituting (2.26) into (2.24) leads to

$$a_r^{\text{new}} = a_{ur}^{\text{new}} - (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \left[C_A (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \right]^{-1} C_A a_{ur}^{\text{new}} \quad (2.27)$$

and the restricted A_r^{new} is obtained from

$$A_r^{\text{new}} = \text{unvec}(a_r^{\text{new}}). \quad (2.28)$$

It remains to optimize the constrained expected conditional log-likelihood with respect to ρ and H . Since the restriction $C_A a = 0$ does not involve ρ and H , maximization with respect to these parameters yields the same formulas as in equations (2.12) and (2.13). The only difference is that the update for H is now evaluated at A_r^{new} , rather than at A_{ur}^{new} . Accordingly, H_r^{new} is now expressed as:

$$H_r^{\text{new}} = \frac{1}{T} \sum_{t=1}^T \left[\Delta y_t \Delta y_t^\top - 2\tilde{R}_t (A_r^{\text{new}})^\top + A_r^{\text{new}} \tilde{S}_t (A_r^{\text{new}})^\top \right]. \quad (2.29)$$

Algorithm 3 summarizes the full EM procedure for the Basic DLM-VECM using A_{ur}^{new} .¹² This procedure requires pre-selecting the cointegration rank r , as well as an initial set of parameters, denoted by:

$$\Theta_{bas}^{(0)} = \{\rho^{(0)}, H^{(0)}, A^{(0)}\} \quad (2.30)$$

¹² If one wishes to estimate A_r^{new} , simply substitute its formula into this algorithm and perform the appropriate substitutions for calculating H_r^{new} .

The specification of r may follow a priori reasoning or standard time series selection methods.¹³ Given the value of r , we recommend setting $H^{(0)}$ and $A^{(0)}$ following the Concentrated Maximum Likelihood procedure of JOHANSEN, and choosing $\rho^{(0)}$ to be very close to one. Furthermore, the selection of the number of lags p ¹⁴ may follow standard information criteria metrics or likelihood ratio tests.

2.3.3 More on the DLM-VECM EM Algorithm: Initialization, Convexity and Overparameterization

Caution must be taken when initializing the parameters of any DLM-VECM EM algorithm. Since the complete log-likelihood, given by equation (2.8), is nonconvex in its parameters, poor initialization may lead to undesirable local maxima. As a practical remedy for this problem, we recommend to try different initializations and/or to adopt initial values supported by theoretical reasoning.

Moreover, the overparameterization arising from the lack of full identification in the decomposition AB_t can complicate the task of overcoming these local maxima through multiple initializations. This is because apparently different parameterizations can correspond to the same likelihood value. Fortunately, the above problem is much less severe when AB_t is unidentified only up to orthogonal rotations, which is the case in the present article.

Accordingly, initializing the parameters following good theoretical reasoning in cases where r is small does not seem to pose a relevant problem. Our practical applications in Chapter 5, for example, employ $r = 2$, which adds only one additional degree of freedom in the DLM-VECM setting.

This concludes the theoretical foundations necessary for modeling the cointegration space through the state vector and the adjustment matrix as a DLM parameter in the Basic DLM-VECM formulation. It remains to extend this framework to the other VECM parameters, namely the intercept, the lag matrices, and the deterministic coefficients. Chapter 3 does so by treating them as DLM parameters, while Chapter 4 treats them as states. In both cases, we will also develop the corresponding EM algorithms.

¹³ Such as the conventional trace or maximum eigenvalue tests of JOHANSEN.

¹⁴ Required only for the DLM-VECM models of Chapters 3 and 4.

Chapter 3

Restricted DLM-VECM

This chapter presents the Restricted DLM-VECM model, which extends the basic DLM-VECM by incorporating the remaining elements of the VECM¹ as exogenous regressors. Consequently, only the cointegration matrix coefficients are specified as time-varying parameters through the state equation. The regression parameters, in turn, are treated as parameters of the DLM model. We begin by formulating the Restricted DLM-VECM model. Then, we develop the associated Restricted DLM-VECM EM algorithm in both its unconstrained and constrained versions.

3.1 Restricted DLM-VECM model

The basic DLM-VECM previously formulated in equations (2.4) and (2.5) is now specified as:

$$\Delta y_t = \tilde{X}_t b_t + Z_t \Psi + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, H), \quad (3.1)$$

$$b_t = \rho b_{t-1} + \eta_t, \quad \eta_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_{nr}), \quad (3.2)$$

where $\tilde{X}_t = (y_{t-1}^\top \otimes A)$ is of dimension $n \times nr$ and A is the static adjustment matrix. Moreover, Z_t is an $n \times n(1 + (p-1) + q)$ exogenous matrix:

$$Z_t = \begin{pmatrix} z_t & 0 & \cdots & 0 \\ 0 & z_t & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & z_t \end{pmatrix}, \quad z_t = (1, \Delta y_{t-1}^\top, \dots, \Delta y_{t-p+1}^\top, d_t^\top), \quad (3.3)$$

b_t is the $nr \times 1$ state vector:

$$b_t = \text{vec}(B_t^\top), \quad (3.4)$$

¹ The intercept, lag matrices, and deterministic coefficients.

and Ψ is the $n(1 + (p - 1) + q) \times 1$ static exogenous vector:

$$\Psi = \text{vec} \left((c, \Gamma_1, \dots, \Gamma_{p-1}, \Phi)^\top \right). \quad (3.5)$$

Additionally, the initial states b_0 follow the distribution given by equation (2.6). The set of parameters associated with the DLM in equations (3.1)–(3.2) is now represented by:

$$\Theta_{res} = \{\rho, H, \Psi, A\} \quad (3.6)$$

Given Θ_{res} , a vector $b_{0|0}$ and the initial state covariance matrix $\Sigma_{0|0}^{(b)} = (1 - \rho^2)^{-1} I_{nr}$, filtering proceeds with the following recursions for $t = 1, \dots, T$:

- **Prediction step:**

$$\begin{aligned} b_{t|t-1} &= \rho b_{t-1|t-1}, \\ \Sigma_{t|t-1}^{(b)} &= \rho^2 \Sigma_{t-1|t-1}^{(b)} + I_{nr}. \end{aligned}$$

- **Update step:**

$$\begin{aligned} \tilde{X}_t &= (y_{t-1}^\top \otimes A) \\ Z_t &= \text{built from } z_t = (1, \Delta y_{t-1}^\top, \dots, \Delta y_{t-p+1}^\top, d_t^\top) \\ b_{t|t} &= b_{t|t-1} + \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top \left(H + \tilde{X}_t \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top \right)^{-1} \left(\Delta y_t - Z_t \Psi - \tilde{X}_t b_{t|t-1} \right), \\ \Sigma_{t|t}^{(b)} &= \Sigma_{t|t-1}^{(b)} - \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top \left(H + \tilde{X}_t \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top \right)^{-1} \tilde{X}_t \Sigma_{t|t-1}^{(b)}. \end{aligned}$$

Algorithm 4 summarizes the filtering procedure for the Restricted DLM-VECM. It outputs $\{b_{t|t-1}, \Sigma_{t|t-1}^{(b)}, b_{t|t}, \Sigma_{t|t}^{(b)}\}_{t=0}^T$, $\{Z_t\}_{t=1}^T$ and $\{\tilde{X}_t\}_{t=1}^T$. The smoothing process, in turn, follows Algorithm 5, yielding $\{b_{t|T}, \Sigma_{t|T}^{(b)}\}_{t=0}^T$ and $\{\Sigma_{t,t-1|T}^{(b)}\}_{t=1}^T$. These smoothed quantities are necessary for updating the parameter set Θ_{res} in the related EM algorithm, which will be presented in the next section.

3.2 Restricted DLM-VECM EM algorithm

The complete-data joint density of the Restricted DLM-VECM² can be written as:

$$p_{\Theta_{res}}(b_{0:T}, \Delta y_{1:T}) = p_{b_{0|0}, \Sigma_{0|0}^{(b)}}(b_0) \prod_{t=1}^T p_\rho(b_t | b_{t-1}) \prod_{t=1}^T p_{A, \Psi, H}(\Delta y_t | b_t), \quad (3.7)$$

where the first term is ruled out, since $b_{0|0}$ and $\Sigma_{0|0}^{(b)}$ are assumed fixed throughout the iterations of the Restricted DLM-VECM EM algorithm. From the Gaussianity assumptions in (2.6), (3.1) and (3.2), the remaining distributions are denoted as $b_t | b_{t-1} \sim \mathcal{N}(\rho b_{t-1}, I_{nr})$

² Equations (2.6), (3.1) and (3.2).

and $\Delta y_t \mid b_t \sim \mathcal{N}(\tilde{X}_t b_t + Z_t \Psi, H)$. Ignoring constants, the complete-data log-likelihood of equation (3.7) is then formulated as

$$\begin{aligned} -2 \ln L(\Theta_{res}) &= \sum_{t=1}^T (b_t - \rho b_{t-1})^\top (b_t - \rho b_{t-1}) + T \ln |H| \\ &\quad + \sum_{t=1}^T (\Delta y_t - \tilde{X}_t b_t - Z_t \Psi)^\top H^{-1} (\Delta y_t - \tilde{X}_t b_t - Z_t \Psi). \end{aligned} \quad (3.8)$$

From this complete-data log-likelihood, we are able to develop both the unconstrained and constrained versions of the EM algorithm for the Restricted DLM-VECM. Since these cases are nearly identical to the Basic DLM-VECM presented in Chapter 2, the rest of this Chapter does not present the derivations in the same level of detail.

3.2.1 Unconstrained EM algorithm

E-step

The expected conditional log-likelihood is now given by:

$$Q(\Theta_{res} \mid \Theta_{res}^{\text{old}}) = \mathbb{E}[-2 \ln L(\Theta_{res}) \mid y_{0:T}, Z_{1:T}, \Theta_{res}^{\text{old}}], \quad (3.9)$$

where $\Theta_{res}^{\text{old}}$ is associated with the previous iteration of the EM procedure. Calculation of the expectations in (3.9) stands for the E-step of the Restricted DLM-VECM EM algorithm:

$$\begin{aligned} Q(\Theta_{res} \mid \Theta_{res}^{\text{old}}) &= \text{tr} \left\{ \sum_{t=1}^T \mathbb{E}[(b_t - \rho b_{t-1})(b_t - \rho b_{t-1})^\top \mid y_{0:T}, Z_{1:T}, \Theta_{res}^{\text{old}}] \right\} + T \ln |H| \\ &\quad + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \mathbb{E}[(\Delta y_t - \tilde{X}_t b_t - Z_t \Psi)(\Delta y_t - \tilde{X}_t b_t - Z_t \Psi)^\top \mid y_{0:T}, Z_{1:T}, \Theta_{res}^{\text{old}}] \right\}. \end{aligned} \quad (3.10)$$

The expectations in the first and second line of (3.10) are denoted by the following expressions:

$$\begin{aligned} \mathbb{E}(b_t \mid y_{0:T}, Z_{1:T}, \Theta_{res}^{\text{old}}) &= b_{t|T}, \\ \mathbb{E}(b_s b_t' \mid y_{0:T}, Z_{1:T}, \Theta_{res}^{\text{old}}) &= b_{s|T} (b_{t|T})^\top + \Sigma_{s,t|T}^{(b)}, \end{aligned}$$

and their evaluation is straightforward. Accordingly, if we define the moment matrices $S_{11}^{(b)}$, $S_{10}^{(b)}$ and $S_{00}^{(b)}$ as in Chapter 2, equation (3.10) may be expressed as:

$$\begin{aligned} Q(\Theta_{res} | \Theta_{res}^{old}) = & \text{tr} \left[S_{11}^{(b)} - \rho S_{10}^{(b)} - \rho S_{10}^{(b)\top} + \rho^2 S_{00}^{(b)} \right] + T \ln |H| \\ & + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \left[\Delta y_t \Delta y_t^\top - 2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top - 2 \Delta y_t \Psi^\top Z_t^\top \right. \right. \\ & \left. \left. + \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top + 2 Z_t \Psi b_{t|T}^\top \tilde{X}_t^\top + Z_t \Psi \Psi^\top Z_t^\top \right] \right\}. \quad (3.11) \end{aligned}$$

M-step

The M-step of the Restricted DLM-VECM EM algorithm updates the parameters ρ , H , Ψ and A . The update formulas for ρ , H and Ψ are given by

$$\rho^{\text{new}} = \left(\text{tr}(S_{10}^{(b)}) \right) \left(\text{tr}(S_{00}^{(b)}) \right)^{-1}, \quad (3.12)$$

$$\begin{aligned} H^{\text{new}} = & \frac{1}{T} \sum_{t=1}^T \left(\Delta y_t \Delta y_t^\top - 2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top - 2 \Delta y_t \Psi^\top Z_t^\top \right. \\ & \left. + 2 \tilde{X}_t b_{t|T} \Psi^\top Z_t^\top + \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top + Z_t \Psi \Psi^\top Z_t^\top \right), \quad (3.13) \end{aligned}$$

$$\Psi^{\text{new}} = \left(\sum_{t=1}^T Z_t^\top (H^{\text{new}})^{-1} Z_t \right)^{-1} \left(\sum_{t=1}^T Z_t^\top (H^{\text{new}})^{-1} (\Delta y_t - \tilde{X}_t b_{t|T}) \right). \quad (3.14)$$

In order to maximize with respect to A , we first apply the identity

$$\tilde{X}_t = (y_{t-1}^\top \otimes A) = A(y_{t-1}^\top \otimes I_r).$$

We then rewrite the terms of equation (3.11) that involve $\tilde{X}_t$ as:

$$\begin{aligned} -2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top &= -2 \Delta y_t b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top A^\top \\ +2 Z_t \Psi b_{t|T}^\top \tilde{X}_t^\top &= +2 Z_t \Psi b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top A^\top \\ \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top &= A (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top A^\top \end{aligned}$$

To simplify our computations, we define

$$\begin{aligned} \tilde{S}_t &= (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top, \\ \tilde{R}_t &= (\Delta y_t - Z_t \Psi) b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top. \end{aligned}$$

Then, the part of (3.11) that depends on A can be written as

$$\begin{aligned} Q(A \mid \Theta_{res}^{old}) &= \sum_{t=1}^T \text{tr} \left[H^{-1} (A \tilde{S}_t A^\top - 2A \tilde{R}_t^\top) \right] + \text{const}, \\ &= \text{tr} \left[H^{-1} (A \tilde{S} A^\top - 2A \tilde{R}^\top) \right] + \text{const}, \\ &= \text{tr} (A \tilde{S} A^\top H^{-1}) - \text{tr} (2A \tilde{R}^\top H^{-1}) + \text{const}, \end{aligned} \quad (3.15)$$

where $\tilde{S} = \sum_{t=1}^T (\tilde{S}_t)$ and $\tilde{R} = \sum_{t=1}^T (\tilde{R}_t)$. An unrestricted maximization of (3.15) with respect to A thus yields :

$$\begin{aligned} A_{ur}^{new} &= \tilde{R} \tilde{S}^{-1}, \\ &= \left(\sum_{t=1}^T (\Delta y_t - Z_t \Psi) b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top \right) \\ &\quad \times \left(\sum_{t=1}^T (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top \right)^{-1}, \end{aligned} \quad (3.16)$$

which is almost identical to the formula presented in equation (2.17). Algorithm 6 summarizes the Unconstrained EM procedure for the Restricted DLM-VECM. Complementarily to the initialization suggestions of Chapter 2, we recommend setting $\Psi^{(0)}$ following the Concentrated Maximum Likelihood procedure of JOHANSEN

3.2.2 Constrained EM algorithm

The Constrained EM algorithm for the Restricted DLM-VECM also imposes $r(r-1)/2$ linear restrictions through $C_A a = 0$. The only difference from the EM algorithm in Chapter 2 is that now we must account for Ψ .

E-step

The expected conditional log-likelihood in equation (3.11) can be rewritten in terms of A as follows:

$$\begin{aligned} Q(\Theta_{res} \mid \Theta_{res}^{old}) &= \text{tr} \left[S_{11}^{(b)} - \rho S_{10}^{(b)} - \rho S_{10}^{(b)\top} + \rho^2 S_{00}^{(b)} \right] + T \ln |H| \\ &\quad + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \left[\Delta y_t \Delta y_t^\top - 2\Delta y_t \Psi^\top Z_t^\top + Z_t \Psi \Psi^\top Z_t^\top - 2\tilde{R}_t A^\top + A \tilde{S}_t A^\top \right] \right\}. \end{aligned} \quad (3.17)$$

This expression can also be reformulated in terms of $a = \text{vec}(A)$, yielding:

$$\begin{aligned} Q(\Theta_{res} | \Theta_{res}^{\text{old}}) = & \text{tr} \left[S_{11}^{(b)} - \rho S_{10}^{(b)} - \rho S_{10}^{(b)\top} + \rho^2 S_{00}^{(b)} \right] + T \ln |H| \\ & + \text{tr} \left\{ H^{-1} \sum_{t=1}^T \left[\Delta y_t \Delta y_t^\top - 2 \Delta y_t \Psi^\top Z_t^\top + Z_t \Psi \Psi^\top Z_t^\top \right] \right\} \\ & - 2a^\top \text{vec}(H^{-1} \tilde{R}) + a^\top (\tilde{S}^\top \otimes H^{-1}) a. \end{aligned} \quad (3.18)$$

M-step

Then, the objective function of the constrained optimization is given by the following Lagrangian:

$$\mathcal{L}(a, \lambda) = a^\top (\tilde{S}^\top \otimes H^{-1}) a - 2a^\top \text{vec}(H^{-1} \tilde{R}) + 2\lambda^\top C_A a, \quad (3.19)$$

where optimization leads to

$$a_r^{\text{new}} = a_{ur}^{\text{new}} - (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \left[C_A (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \right]^{-1} C_A a_{ur}^{\text{new}}. \quad (3.20)$$

The restricted A_r^{new} is obtained from $\text{unvec}(a_r^{\text{new}})$. Again, the update formulas for ρ, H, Ψ in the constrained EM algorithm are given by equations (3.12), (3.13), and (3.14), respectively. The only difference is that H and Ψ are evaluated at the restricted estimator A_r^{new} instead of A_{ur}^{new} .

Chapter 4

General DLM-VECM

This chapter develops the General DLM-VECM. This model extends the Restricted DLM-VECM from Chapter 3 by incorporating the exogenous parameters into the state equation. After presenting the General DLM-VECM, we provide the related General DLM-VECM EM algorithm in both its unconstrained and constrained versions.

4.1 General DLM-VECM model

The General DLM-VECM incorporates the exogenous coefficients Ψ from the Restricted DLM-VECM into the state equation. Accordingly, these parameters become time-varying and are now denoted by Ψ_t . Thus, the state vector is now represented by:

$$\theta_t = \left(b_t^\top \Psi_t^\top \right)^\top, \quad (4.1)$$

where

$$\Psi_t = \text{vec} \left(\left(c_t, \Gamma_{t,1}, \dots, \Gamma_{t,p-1}, \Phi_t \right)^\top \right), \quad (4.2)$$

and $\dim(\theta_t) = nr + n(1 + (p-1) + q) = nr + k$. The General DLM-VECM is then specified as:

$$\Delta y_t = F_t \theta_t + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, H), \quad (4.3)$$

$$\theta_t = G \theta_{t-1} + \eta_t, \quad \eta_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, Q), \quad (4.4)$$

where the measurement matrix F_t and the transition matrix G are given by

$$F_t = \begin{pmatrix} \tilde{X}_t & Z_t \end{pmatrix}, \quad (4.5)$$

$$G = \begin{pmatrix} \rho I_{nr} & 0 \\ 0 & I_k \end{pmatrix}. \quad (4.6)$$

Additionally, the state innovation covariance matrix is denoted by

$$Q = \begin{pmatrix} I_{nr} & 0 \\ 0 & Q^{(\Psi)} \end{pmatrix}, \quad (4.7)$$

so that $\eta_t = (\eta_t^{(b)\top}, \eta_t^{(\Psi)\top})^\top$. Moreover, b_t , $\tilde{X}_t$, and Z_t are defined equivalently as in Chapter 3. The initial state vector $\theta_0 = (b_0^\top \Psi_0^\top)^\top$ has the following distribution:

$$\theta_0 \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ \Psi_{0|0} \end{pmatrix}, \begin{pmatrix} (1 - \rho^2)^{-1} I_{nr} & 0 \\ 0 & \Sigma_{0|0}^{(\Psi)} \end{pmatrix}\right), \quad (4.8)$$

From (4.8), $b_{0|0}$ can either be sampled from $\mathcal{N}(0, (1 - \rho^2)^{-1} I_{nr})$ or set to suitable initial values, while $\Sigma_{0|0}^{(b)}$ is set to be $(1 - \rho^2)^{-1} I_{nr}$. Moreover, both $b_{0|0}$ and $\Sigma_{0|0}^{(b)}$ are considered fixed and not updated within the associated EM procedure. In contrast, $\Psi_{0|0}$ and $\Sigma_{0|0}^{(\Psi)}$ are treated as parameters of the General DLM-VECM.

Accordingly, the set of parameters associated with the DLM in equations (4.3)–(4.8) is represented by

$$\Theta_{gen} = \{\Psi_{0|0}, \Sigma_{0|0}^{(\Psi)}, \rho, H, Q^{(\Psi)}, A\}. \quad (4.9)$$

Given Θ_{gen} , $b_{0|0}$, $\Sigma_{0|0}^{(b)}$, and the initial parameters $\Psi_{0|0}$ and $\Sigma_{0|0}^{(\Psi)}$, filtering proceeds with the following recursions for $t = 1, \dots, T$:

- **Prediction step:**

$$\begin{aligned} \theta_{t|t-1} &= G\theta_{t-1|t-1}, \\ \Sigma_{t|t-1} &= G\Sigma_{t-1|t-1}G^\top + Q, \end{aligned}$$

- **Update step:**

$$\begin{aligned} \tilde{X}_t &= (y_{t-1}^\top \otimes A), \\ Z_t &= \text{built from } z_t = (1, \Delta y_{t-1}^\top, \dots, \Delta y_{t-p+1}^\top, d_t^\top), \\ F_t &= (\tilde{X}_t \quad Z_t). \\ \theta_{t|t} &= \theta_{t|t-1} + \Sigma_{t|t-1} F_t^\top \left(H + F_t \Sigma_{t|t-1} F_t^\top \right)^{-1} (\Delta y_t - F_t \theta_{t|t-1}), \\ \Sigma_{t|t} &= \Sigma_{t|t-1} - \Sigma_{t|t-1} F_t^\top \left(H + F_t \Sigma_{t|t-1} F_t^\top \right)^{-1} F_t \Sigma_{t|t-1}. \end{aligned}$$

Algorithm 7 summarizes the filtering procedure for the General DLM-VECM. It outputs $\{\theta_{t|t-1}, \Sigma_{t|t-1}, \theta_{t|t}, \Sigma_{t|t}\}_{t=0}^T$ and $\{F_t\}_{t=1}^T$. Accordingly, the smoothing process follows algorithm 8, yielding $\{\theta_{t|T}, \Sigma_{t|T}\}_{t=0}^T$ and $\{\Sigma_{t,t-1|T}\}_{t=1}^T$. These quantities are required for updating Θ_{gen} in the EM algorithm, which we now present.

4.2 General DLM-VECM EM algorithm

We begin by writing the complete-data joint density of the General DLM-VECM EM algorithm, which is formulated as:

$$p_{\Theta_{gen}}(\theta_{0:T}, \Delta y_{1:T}) = p_{\Psi_{0|0}, \Sigma_{0|0}^{(\Psi)}}(\theta_0) \prod_{t=1}^T p_{\rho, Q^{(\Psi)}}(\theta_t | \theta_{t-1}) \prod_{t=1}^T p_{A, H}(\Delta y_t | \theta_t), \quad (4.10)$$

Moreover, the above distributions are specified as

- $\theta_0 \sim \mathcal{N}(\theta_{0|0}, \Sigma_{0|0})$,
- $\theta_t | \theta_{t-1} \sim \mathcal{N}(G\theta_{t-1}, Q)$,
- $\Delta y_t | \theta_t \sim \mathcal{N}(F_t\theta_t, H)$,

with

$$\begin{aligned} \theta_{0|0} &= \begin{pmatrix} 0 \\ \Psi_{0|0} \end{pmatrix}, \quad \Sigma_{0|0} = \begin{pmatrix} (1 - (\rho_{init})^2)^{-1} I_{nr} & 0 \\ 0 & \Sigma_{0|0}^{(\Psi)} \end{pmatrix}, \quad G = \begin{pmatrix} \rho I_{nr} & 0 \\ 0 & I_k \end{pmatrix}, \\ Q &= \begin{pmatrix} I_{nr} & 0 \\ 0 & Q^{(\Psi)} \end{pmatrix}, \quad F_t = (\tilde{X}_t \ Z_t), \quad \tilde{X}_t = (y_{t-1}^\top \otimes A). \end{aligned} \quad (4.11)$$

In the above matrix $\Sigma_{0|0}$, the scalar ρ_{init} denotes a fixed initial value for the autoregressive coefficient. In contrast, the ρ appearing in G is treated as a parameter to be updated through the actual EM algorithm. This separation between ρ_{init} and ρ is a very convenient choice that leads to analytical solutions for ρ in the M-step.

4.2.1 Unconstrained EM algorithm

E-step

From the expected conditional log-likelihood:

$$Q(\Theta_{gen} | \Theta_{gen}^{old}) = \mathbb{E}[-2 \ln L(\Theta_{gen}) | y_{0:T}, F_{1:T}, \Theta_{gen}^{old}], \quad (4.12)$$

the E-step of the General DLM-VECM is now given by:

$$\begin{aligned} Q(\Theta_{gen} | \Theta_{gen}^{old}) &= \ln |\Sigma_{0|0}| + \mathbb{E}[(\theta_0 - \theta_{0|0})^\top \Sigma_{0|0}^{-1} (\theta_0 - \theta_{0|0}) | y_{0:T}, F_{1:T}, \Theta_{gen}^{old}] \\ &\quad + T \ln |Q| + \sum_{t=1}^T \mathbb{E}[(\theta_t - G\theta_{t-1})^\top Q^{-1} (\theta_t - G\theta_{t-1}) | y_{0:T}, F_{1:T}, \Theta_{gen}^{old}] \\ &\quad + T \ln |H| + \sum_{t=1}^T \mathbb{E}[(\Delta y_t - F_t\theta_t)^\top H^{-1} (\Delta y_t - F_t\theta_t) | y_{0:T}, F_{1:T}, \Theta_{gen}^{old}]. \end{aligned} \quad (4.13)$$

The expectations in (4.13) are associated with the following expressions:

$$\begin{aligned}\mathbb{E}(\theta_t \mid y_{0:T}, Z_{1:T}, \Theta_{gen}^{old}) &= \theta_{t|T}, \\ \mathbb{E}(\theta_s \theta'_t \mid y_{0:T}, Z_{1:T}, \Theta_{gen}^{old}) &= \theta_{s|T}(\theta_{t|T})^\top + \Sigma_{s,t|T}^{(\theta)},\end{aligned}$$

Moreover, by defining the moment matrices

$$\begin{aligned}S_{11}^{(\theta)} &= \sum_{t=1}^T \left(\theta_{t|T}(\theta_{t|T})^\top + \Sigma_{t|T}^{(\theta)} \right), \\ S_{10}^{(\theta)} &= \sum_{t=1}^T \left(\theta_{t|T}(\theta_{t-1|T})^\top + \Sigma_{t,t-1|T}^{(\theta)} \right), \\ S_{00}^{(\theta)} &= \sum_{t=1}^T \left(\theta_{t-1|T}(\theta_{t-1|T})^\top + \Sigma_{t-1|T}^{(\theta)} \right),\end{aligned}$$

equation (4.13) is simplified to

$$\begin{aligned}Q(\Theta_{gen} \mid \Theta_{gen}^{old}) &= \ln |\Sigma_{0|0}| + \text{tr} \left[\Sigma_{0|0}^{-1} (\theta_{0|T} \theta_{0|T}^\top + \Sigma_{0|T}^{(\theta)} - 2\theta_{0|T} \theta_{0|0}^\top + \theta_{0|0} \theta_{0|0}^\top) \right] \\ &+ T \ln |Q| + \text{tr} \left[Q^{-1} (S_{11}^{(\theta)} - G S_{10}^{(\theta)\top} - S_{10}^{(\theta)} G^\top + G S_{00}^{(\theta)} G^\top) \right] \\ &+ T \ln |H| + \sum_{t=1}^T \text{tr} \left[H^{-1} (\Delta y_t \Delta y_t^\top - 2\Delta y_t \theta_{t|T}^\top F_t^\top + F_t (\theta_{t|T} \theta_{t|T}^\top + \Sigma_{t|T}^{(\theta)}) F_t^\top) \right].\end{aligned}\quad (4.14)$$

M-step

Before proceeding with the optimization, we introduce some notation that will facilitate the understanding of the derivations. To this end, we denote the moment matrices and the smoothed covariance matrix by a block decomposition that explicitly shows the separation between the state parameters b_t and Ψ_t , that is,

$$S_{ij}^{(\theta)} = \begin{pmatrix} S_{ij}^{(b)} & S_{ij}^{(b,\Psi)} \\ S_{ij}^{(\Psi,b)} & S_{ij}^{(\Psi)} \end{pmatrix}, \quad \Sigma_{t|T}^{(\theta)} = \begin{pmatrix} \Sigma_{t|T}^{(b)} & \Sigma_{t|T}^{(b,\Psi)} \\ \Sigma_{t|T}^{(\Psi,b)} & \Sigma_{t|T}^{(\Psi)} \end{pmatrix}. \quad (4.15)$$

Moreover, we highlight that the inverse of the state covariance matrix is also diagonal, being expressed as

$$Q^{-1} = \begin{pmatrix} I_{nr} & 0 \\ 0 & (Q^{(\Psi)})^{-1} \end{pmatrix}. \quad (4.16)$$

Minimization of (4.14) with respect to Θ_{gen} corresponds to the M-step of the General DLM-VECM EM algorithm, where some of the matrices appearing in (4.14) are linked to the model parameters through the definitions in (4.11). Accordingly, given the smoothed state estimates, the system parameters are updated. First optimizing with respect to $\Psi_{0|0}$

and $\Sigma_{0|0}^{(\Psi)}$ yields:

$$\Psi_{0|0}^{\text{new}} = \begin{pmatrix} 0_{k \times nr} & I_k \end{pmatrix} \theta_{0|T}, \quad (4.17)$$

$$\Sigma_{0|0}^{(\Psi)\text{new}} = \begin{pmatrix} 0_{k \times nr} & I_k \end{pmatrix} \Sigma_{0|T}^{(\theta)} \begin{pmatrix} 0_{k \times nr} \\ I_k \end{pmatrix}. \quad (4.18)$$

Then, minimizing with respect to the state equation parameters leads to

$$\rho^{\text{new}} = \frac{\text{tr}(S_{10}^{(b)})}{\text{tr}(S_{00}^{(b)})}, \quad (4.19)$$

$$Q^{(\Psi)\text{new}} = \frac{1}{T} \left(S_{11}^{(\Psi)} - S_{10}^{(\Psi)} - S_{10}^{(\Psi)\top} + S_{00}^{(\Psi)} \right). \quad (4.20)$$

Finally, we must update H and A with similar formulas as in equations (3.13) and (3.16). Updating H is straightforward and yields

$$H^{\text{new}} = \frac{1}{T} \sum_{t=1}^T \left(\Delta y_t \Delta y_t^\top - 2 \Delta y_t \theta_{t|T}^\top F_t^\top + F_t (\theta_{t|T} \theta_{t|T}^\top + \Sigma_{t|T}^{(\theta)}) F_t^\top \right) \quad (4.21)$$

However, the update of A presents more complicated derivations. Given that

$$F_t = (\tilde{X}_t \ Z_t), \quad \tilde{X}_t = A(y_{t-1}^\top \otimes I_r), \quad \theta_t = \begin{pmatrix} b_t \\ \Psi_t \end{pmatrix},$$

the terms of (4.14) that involve F_t are denoted by the following linear and quadratic terms

- Linear:

$$-2 \Delta y_t \theta_{t|T}^\top F_t^\top = -2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top - 2 \Delta y_t \Psi_{t|T}^\top Z_t^\top,$$

- Quadratic:

$$\begin{aligned} F_t \theta_{t|T} \theta_{t|T}^\top F_t^\top &= \tilde{X}_t b_{t|T} b_{t|T}^\top \tilde{X}_t^\top + \tilde{X}_t b_{t|T} \Psi_{t|T}^\top Z_t^\top + Z_t \Psi_{t|T} b_{t|T}^\top \tilde{X}_t^\top + Z_t \Psi_{t|T} \Psi_{t|T}^\top Z_t^\top, \\ F_t \Sigma_{t|T}^{(\theta)} F_t^\top &= \tilde{X}_t \Sigma_{t|T}^{(b)} \tilde{X}_t^\top + \tilde{X}_t \Sigma_{t|T}^{(b,\Psi)} Z_t^\top + Z_t \Sigma_{t|T}^{(\Psi,b)} \tilde{X}_t^\top + Z_t \Sigma_{t|T}^{(\Psi)} Z_t^\top. \end{aligned}$$

Now, ignoring the terms that do not involve $\tilde{X}_t$ and substituting $\tilde{X}_t = A(y_{t-1}^\top \otimes I_r)$, the A -dependent components of (4.14) reduce to

$$\begin{aligned} -2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top &= -2 \Delta y_t b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top A^\top, \\ \tilde{X}_t b_{t|T} b_{t|T}^\top \tilde{X}_t^\top &= A(y_{t-1}^\top \otimes I_r) b_{t|T} b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top A^\top, \\ \tilde{X}_t \Sigma_{t|T}^{(b)} \tilde{X}_t^\top &= A(y_{t-1}^\top \otimes I_r) \Sigma_{t|T}^{(b)} (y_{t-1}^\top \otimes I_r)^\top A^\top, \\ \tilde{X}_t b_{t|T} \Psi_{t|T}^\top Z_t^\top &= A(y_{t-1}^\top \otimes I_r) b_{t|T} \Psi_{t|T}^\top Z_t^\top, \\ Z_t \Psi_{t|T} b_{t|T}^\top \tilde{X}_t^\top &= Z_t \Psi_{t|T} b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top A^\top, \\ \tilde{X}_t \Sigma_{t|T}^{(b,\Psi)} Z_t^\top &= A(y_{t-1}^\top \otimes I_r) \Sigma_{t|T}^{(b,\Psi)} Z_t^\top, \\ Z_t \Sigma_{t|T}^{(\Psi,b)} \tilde{X}_t^\top &= Z_t \Sigma_{t|T}^{(\Psi,b)} (y_{t-1}^\top \otimes I_r)^\top A^\top. \end{aligned}$$

Then, collecting the terms that multiply $\tilde{X}_t$ and using the properties of the trace operator, we define the matrices

$$\tilde{S}_t = (y_{t-1}^\top \otimes I_r)(b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)})(y_{t-1}^\top \otimes I_r)^\top, \quad (4.22)$$

$$\tilde{R}_t = \left(\Delta y_t b_{t|T}^\top - Z_t \Psi_{t|T} b_{t|T}^\top - Z_t \Sigma_{t|T}^{(\Psi, b)} \right) (y_{t-1}^\top \otimes I_r)^\top. \quad (4.23)$$

This allows rewriting the part of (4.14) that depends on A as

$$\begin{aligned} Q(A \mid \Theta_{gen}^{old}) &= \sum_{t=1}^T \text{tr} \left[H^{-1} (A \tilde{S}_t A^\top - 2A \tilde{R}_t^\top) \right] + \text{const}, \\ &= \text{tr} \left[H^{-1} (A \tilde{S} A^\top - 2A \tilde{R}^\top) \right] + \text{const}, \\ &= \text{tr} (A \tilde{S} A^\top H^{-1}) - \text{tr} (2A \tilde{R}^\top H^{-1}) + \text{const}, \end{aligned} \quad (4.24)$$

where $\tilde{S} = \sum_{t=1}^T (\tilde{S}_t)$ and $\tilde{R} = \sum_{t=1}^T (\tilde{R}_t)$. Therefore, maximization in the M-step with respect to A yields

$$\begin{aligned} A_{ur}^{new} &= \tilde{R} \tilde{S}^{-1}, \\ &= \left(\sum_{t=1}^T (\Delta y_t b_{t|T}^\top - Z_t \Psi_{t|T} b_{t|T}^\top - Z_t \Sigma_{t|T}^{(\Psi, b)}) (y_{t-1}^\top \otimes I_r)^\top \right) \\ &\quad \times \left(\sum_{t=1}^T (y_{t-1}^\top \otimes I_r)(b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)})(y_{t-1}^\top \otimes I_r)^\top \right)^{-1}. \end{aligned} \quad (4.25)$$

Algorithm 9 summarizes the complete EM procedure for the General DLM-VECM. The new initialization parameters $\Psi_{0|0}^{(0)}$, $\Sigma_{0|0}^{(\Psi)(0)}$, and $Q^{(\Psi)(0)}$ may be chosen according to the uninformative case, in which $\Sigma_{0|0}^{(\Psi)(0)}$ and $Q^{(\Psi)(0)}$ are set to large diagonal matrices and $\Psi_{0|0}^{(0)}$ is sampled from $\mathcal{N} = (0, \Sigma_{0|0}^{(\Psi)(0)})$. Alternatively, they may be set to suitable initial values that follow maximum likelihood estimation procedures.

4.2.2 Constrained EM algorithm

The Constrained EM algorithm for the General DLM-VECM follows the same rationale as those presented in Chapters 2 and 3. The only difference is that we must incorporate the new definition of $\tilde{R}_t$ given in equation (4.23). Accordingly, optimization leads to

$$a_r^{new} = a_{ur}^{new} - (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \left[C_A (\tilde{S}^\top \otimes H^{-1})^{-1} C_A^\top \right]^{-1} C_A a_{ur}^{new}, \quad (4.26)$$

where the restricted A_r^{new} is obtained from $\text{unvec}(a_r^{new})$. The update formulas for $\Psi_{0|0}, \Sigma_{0|0}^{(\Psi)}, \rho, H$ and $Q^{(\Psi)}$ in the constrained EM algorithm are given by equations (4.17), (4.18), (4.19), (4.20) and (4.21), respectively. Again, the only difference is that H is evaluated at the restricted estimator A_r^{new} instead of A_{ur}^{new} .

Chapter 5

Empirical Application: Fuel Dataset

This chapter applies the DLM-VECM framework to a weekly dataset of prices in the Brazilian retail fuel market. We begin by presenting the data. Then, we analyze the evolution of the cointegration space under the Restricted DLM-VECM and the General DLM-VECM. Finally, we compare the predictive performance of the proposed DLM-VECM models along with some baseline alternatives across different forecasting horizons.

5.1 Data

Our dataset comprises weekly data on gasoline, diesel, and ethanol prices in the Brazilian retail fuel market from May 2004 to November 2025. Specifically, we use the weekly average retail price computed by the Brazilian National Agency for Petroleum, Natural Gas and Biofuels (ANP), which is obtained from a large and geographically diverse sample of fuel stations across Brazil.¹

Figure 5.1 displays the whole sample for the three price series, where the dashed vertical line at 2020-01-01 corresponds to the separation date between training and test sets. Moreover, a small number of observations are missing from the series,² and they are filled using cubic spline interpolation.

Throughout the training sample, all price series appear to share one common stochastic trend. In the present case with $n = 3$, a single stochastic trend corresponds to 2 cointegration relationships, which leads to choosing $r = 2$ for our practical applications. In this context, the unnormalized cointegration matrix at time t would have the following form:

$$B_t = \begin{pmatrix} b_{11,t} & b_{12,t} \\ b_{21,t} & b_{22,t} \\ b_{31,t} & b_{32,t} \end{pmatrix}. \quad (5.1)$$

¹ Information about the methodology of data collection can be found online at <https://www.gov.br/anp/pt-br/assuntos/precos-e-defesa-da-concorrencia/precos>.

² Dates: 2005-08-21, 2005-08-28, 2009-08-23, 2009-08-30, 2015-08-23, 2020-08-30, 2020-09-06, 2020-09-13, 2020-09-20, 2020-09-27, 2020-10-04, 2020-10-11, and 2020-10-18.

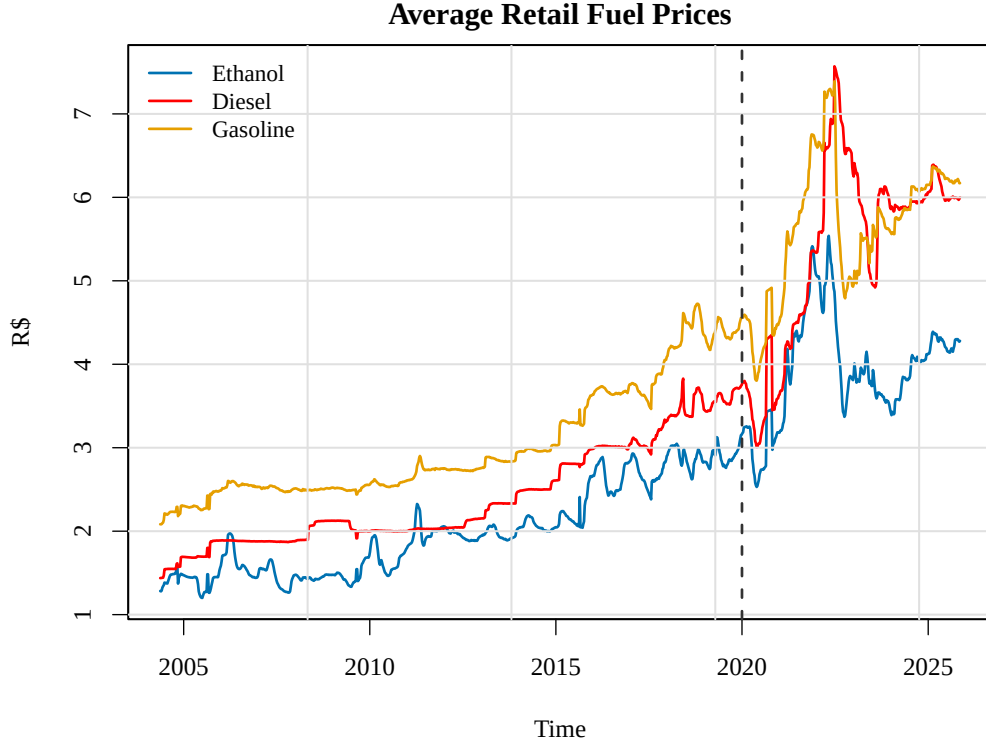


Figure 5.1: Average Retail Fuel Prices - Brazil.

However, the matrix representation in (5.1) is not uniquely identified. We are only able to identify the column space of matrix B_t , that is, the cointegration space. To ensure interpretability, we adopt the linear normalization of JOHANSEN in order to analyze the evolution of the cointegration space.³ This identification scheme places r^2 identifying restrictions on B_t , transforming the unnormalized matrix in (5.1) into

$$B_t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ b_t^{(\text{eth})} & b_t^{(\text{die})} \end{pmatrix}, \quad (5.2)$$

where the variables are ordered as (ethanol, diesel, gasoline). Since the linear normalization uniquely identifies the cointegration space, the constrained versions of the EM algorithm presented in Chapters 3 and 4 are not required for our empirical analysis. As already discussed, these procedures would be useful in the case of severe overparameterization. Since $r = 2$ adds only one degree of freedom, such additional restrictions do not seem necessary.

The coefficients in (5.2) imply that the ratios $1 : b_t^{(\text{eth})}$ between ethanol and gasoline, and $1 : b_t^{(\text{die})}$ between diesel and gasoline, are stationary. In this context, curiosity arises in the following question: what are the fundamentals behind each one of these cointegration relationships?

The case of ethanol and gasoline is supported by a clear structural interpretation. Since

³ Section 5.2.

Brazil possesses a large fleet of flex-fuel vehicles, ethanol and gasoline are substitute goods in a microeconomic sense. In this context, their price adjustment ratio is mainly driven by the relative energy efficiency of the two fuels. Ask any flex-vehicle owner what $b_t^{(eth)}$ should be, and they will answer with conviction: "It should be 70%. If the ethanol price is higher than 70% of the gasoline price, I fill up with gasoline; otherwise, I use ethanol.". In contrast, the stationarity between diesel and gasoline is related to the fact that both fuels are products of crude oil refining, thereby sharing common cost components.

However, these cointegration relationships are subject to further distortions. Historically, the Brazilian retail fuel market has been strongly influenced by institutional factors, such as taxation and fuel pricing policies, which have led to different pricing regimes over time. In this context, our DLM-VECM formulations may be able to capture these regime changes automatically; therefore employing one of our models can bring additional gains relative to its competitors. With this in mind, we now proceed with the empirical application of the DLM-VECM framework to the fuel price dataset.

Section 5.2 analyzes the evolution of the cointegration space under the Restricted DLM-VECM and the General DLM-VECM, based on the smoothed states. While the Restricted DLM-VECM displays a cointegration space with economic meaning, the General DLM-VECM does not. The main reason for this is that the General DLM-VECM is excessively flexible and prone to overfitting. Furthermore, Section 5.4 compares the predictive performance of our DLM-VECM proposals with standard alternatives using the mean squared forecast errors (MSFE).

5.2 Evolution of the Cointegration Space

5.2.1 Restricted DLM-VECM

This subsection analyzes the evolution of the cointegration space over time under the Restricted DLM-VECM for $r = 2$ cointegration relationships. To this end, we must select convenient input values for p and Θ_{res}^0 . First, the choice $p = 3$ is determined via a likelihood ratio test. Moreover, $\rho^{(0)} \in \Theta_{res}^{(0)}$ is set to 0.999, while the remaining parameters of Θ_{res}^0 come from the maximum likelihood estimation procedure of JOHANSEN.

After this initialization step, deriving the cointegration spaces is straightforward. We begin by estimating the parameter set $\hat{\Theta}_{res}$ using Algorithm 6. We then compute the smoothed states by applying Algorithm 5. Finally, the smoothed states are normalized according to equation (5.2), yielding the coefficients $b_{t|T}^{(eth)}$ and $b_{t|T}^{(die)}$, which are shown in Figure 5.2. In the referred figure, the dashed blue line corresponds to the horizontal line $y = 0.7$, and the dashed gray vertical line corresponds to the date 2020-01-01, which separates the train and test sets. Accordingly, $\hat{\Theta}_{res}$ is estimated using the training set, and Figure 5.2 exhibits the cointegration space both in-sample and out-of-sample.

Notice how $b_{t|T}^{(eth)}$ has oscillated around the magic ratio of 0.7 most of the time, especially from 2010 onward. Depending on the period, deviations from the dashed blue line may be linked to different pricing policies for ethanol and gasoline. For example, gasoline prices were controlled during certain periods to contain inflation or were adjusted less frequently relative to international benchmarks, while ethanol prices were more market-driven.

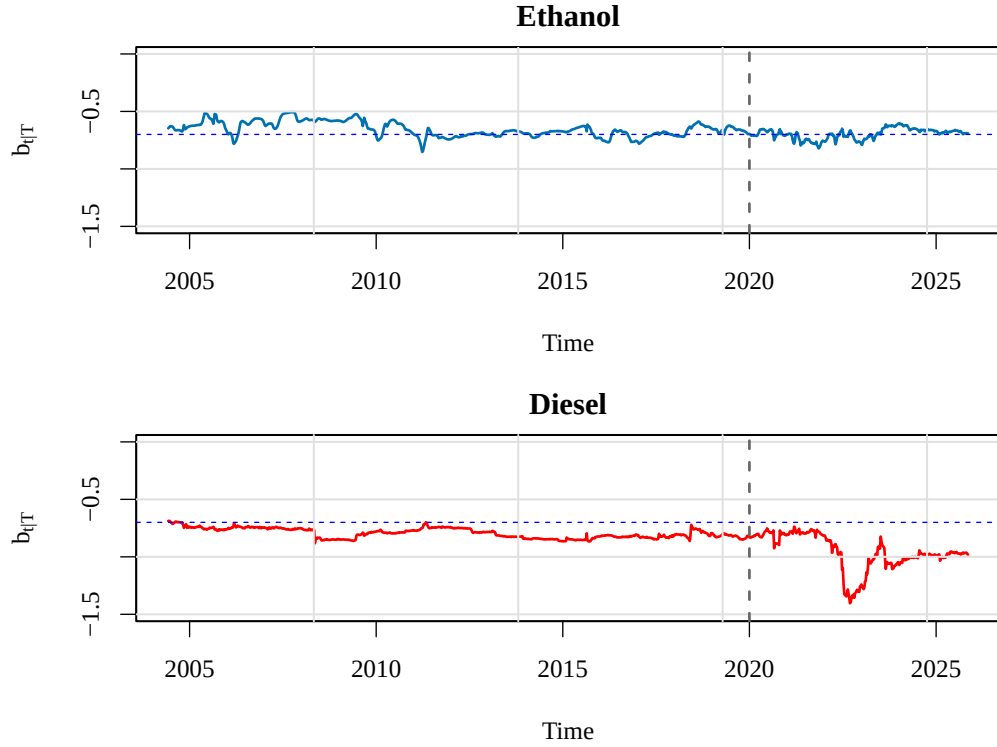


Figure 5.2: *Restricted DLM-VECM: Normalized Smoothed coefficients for $r = 2$.*

With respect to diesel prices, they were also subject to strong government pricing policies until 2016. Notice how $b_{t|T}^{(die)}$ exhibits a clear ratio in the beginning of the sample, which disentangles in more recent years. For example, the fall of $b_{t|T}^{(die)}$ around 2022 is linked to an exchange rate depreciation and external supply shocks, along with the higher relative dependence of diesel on imports with respect to gasoline. Moreover, from 2023 onward, the International Price Parity policy was discontinued, resulting in a more stable value for $b_{t|T}^{(die)}$. These scenarios exemplify the capability of our model to capture structural breaks and time-varying behavior in cointegration relationships.

Additionally, we can also analyze the evolution of the cointegration space under $r = 1$. In this case, the normalized cointegration matrix B_t is given by

$$B_t = \begin{pmatrix} 1 \\ b_t^{(die)} \\ b_t^{(gas)} \end{pmatrix}, \quad (5.3)$$

where the linear combination $1 \times \text{Ethanol} + b_t^{(die)} \times \text{Diesel} + b_t^{(gas)} \times \text{Gasoline}$ is assumed stationary. Please note that the superscript interpretation of $b_t^{(die)}$ and $b_t^{(gas)}$ differs from that of the coefficients in equation (5.2). Figure 5.3 displays the evolution of the cointegration space following this coefficient definition.

From this plot, we infer that ethanol plays a negligible role in the cointegration relationship up to 2016–2017. Subsequently, it appears to enter this relationship more strongly in recent years. Moreover, notice how the normalized coefficients for diesel and gasoline

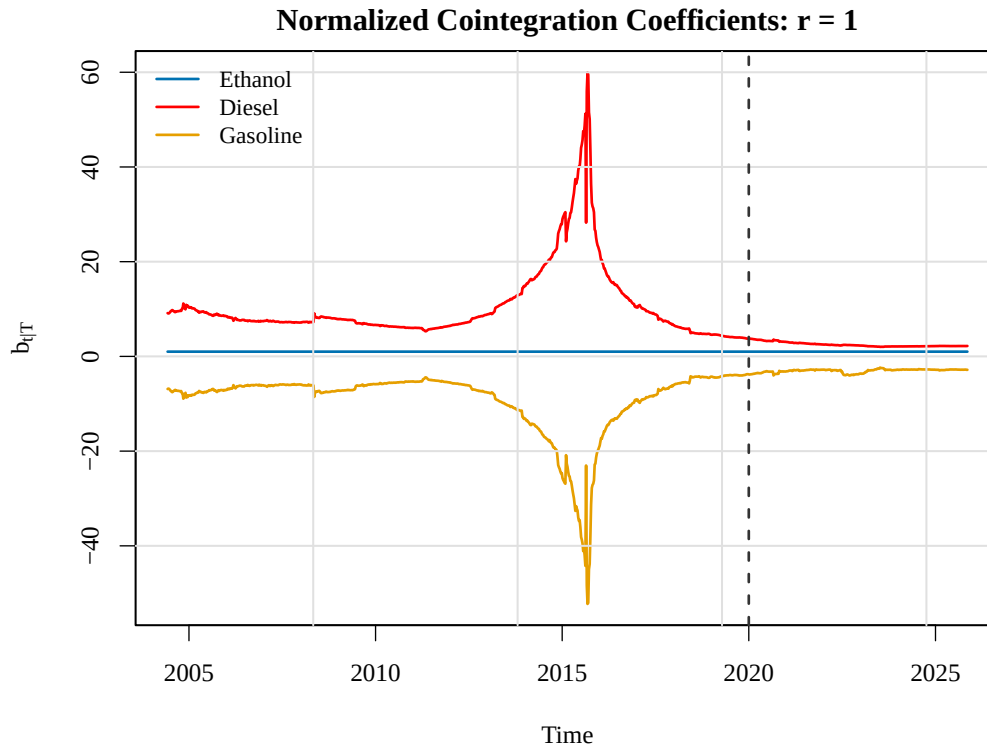


Figure 5.3: *Restricted DLM-VECM: Normalized Smoothed coefficients for $r = 1$.*

display a somewhat mirrored trajectory, diverging only in magnitude.

5.2.2 General DLM-VECM

To analyze the evolution of the cointegration space under the General DLM-VECM, we set $r = 2$, $p = 3$, $\rho^0 = 0.999$, and choose the remaining elements of Θ_{gen}^0 according to the maximum likelihood estimation procedure of JOHANSEN. Moreover, we follow the same methods of the previous subsection. Figure 5.4 below displays the same coefficients of Figure 5.2, but for the General DLM-VECM. Notice that the time evolution of these coefficients does not possess a clear structural interpretation over time, in contrast to the Restricted DLM-VECM.

The reason for this behavior emerges from the excessive flexibility of the General DLM-VECM. For $r = 2$ and $p = 3$, this model results in 27 VECM regression parameters that vary over time without restrictions. Moreover, the block $Q^{(\Psi)}$ of the state covariance matrix has "only" $\frac{(21+1)(21)}{2} = 231$ free parameters. Accordingly, it is not surprising that the cointegration space does not exhibit clear economic meaning and that its predictive performance in Section 5.4 is poor: the General DLM-VECM in its baseline form severely overfits.

5.3 Improving the General DLM-VECM

Before proceeding to the forecasting comparison results, we make a few suggestions for improvements to the General DLM-VECM. Since implementing and verifying these

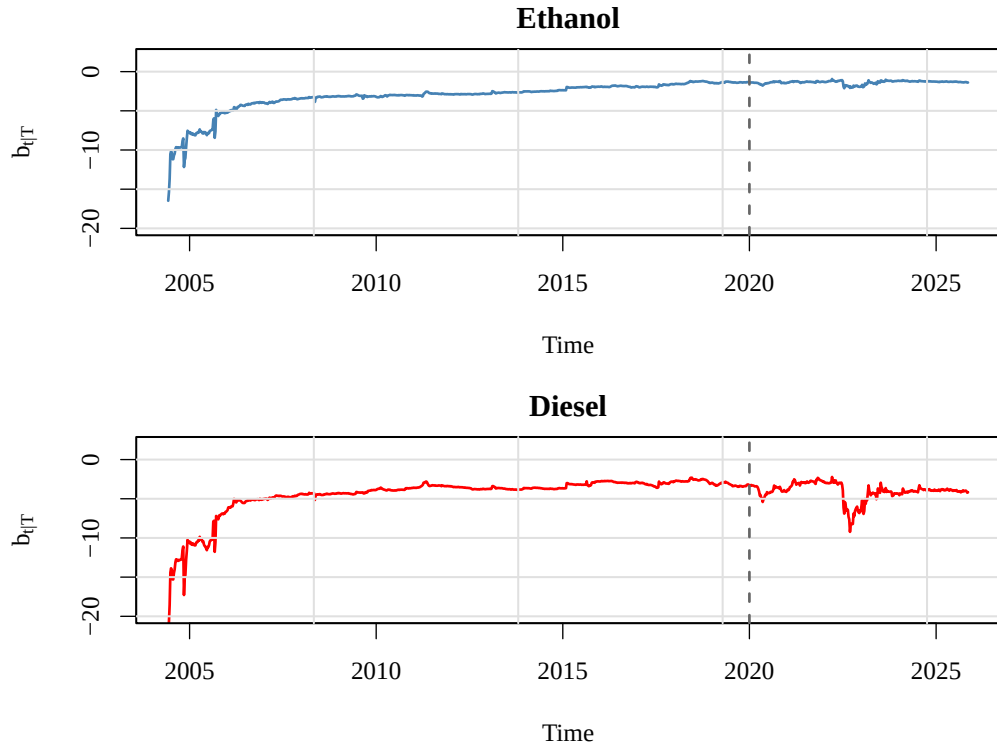


Figure 5.4: General DLM-VECM: Normalized Smoothed coefficients for $r = 2$.

suggestions would require a new study, we limit ourselves to a simple discussion of possible directions.

We recommend setting restrictions on both the parameter set Θ_{gen} and the states. Restrictions on the parameter set are easily implemented by maximizing the expected conditional log-likelihood in the M-step subject to linear constraints. For example, one may impose zero-restrictions on $Q^{(\Psi)}$ or H in order to obtain diagonal/block-diagonal matrices.

On the other hand, restrictions on the states are implementable by two strategies. In the first strategy, we proceed by incorporating an exogenous matrix Z_t , as in equation (3.3), corresponding to a subset of static parameters. For example, one may treat the intercepts and cross-difference parameters as fixed over time, thereby updating them through the M-step of the EM algorithm. Accordingly, even shrinkage through Lasso/Ridge methods may be introduced within this approach.

In the second strategy, we specify the initial distribution of the states as an informative prior for the future states.⁴ In this context, the Minnesota prior of LITTELMAN, 1986 can be introduced, as in KOOP and KOROBILIS, 2013. Furthermore, this type of strategy can even allow informative priors on the cointegration space.

⁴ To preserve the structure, $\theta_{0|0}$ and $\Sigma_{0|0}$ are not updated within the M-step.

5.4 Forecasting Comparison

Using 2020-01-01 as the separation date between the training and test samples, we are left with 306 observations to assess the predictive performance of the following models on our fuel price dataset:

1. A Restricted DLM-VECM with $r = 1$.
2. A Restricted DLM-VECM with $r = 2$.
3. A Restricted DLM-VECM with $r = 3$.⁵
4. A General DLM-VECM with $r = 2$.
5. A Random Walk.
6. A static VAR in levels.
7. A static VECM.

All models, except for the random walk, adopt a lag order of $p = 3$. We compare their performance across forecast horizons of $h = 1, \dots, 6$ weeks using the mean squared forecast error (MSFE). Tables 5.1 to 5.6 below report the MSFE for the referred models for these horizons.

	Res _{$r=1$}	Res _{$r=2$}	Res _{$r=3$}	Gen _{$r=2$}	RW	VAR	VECM
Ethanol	0.0074	0.0068	0.0068	0.0203	0.0070	0.0090	0.0090
Diesel	0.0138	0.0138	0.0137	0.3474	0.0141	0.0140	0.0140
Gasoline	0.0093	0.0090	0.0090	0.0211	0.0096	0.0103	0.0103

Table 5.1: MSFE for $h = 1$.

	Res _{$r=1$}	Res _{$r=2$}	Res _{$r=3$}	Gen _{$r=2$}	RW	VAR	VECM _{$r=2$}
Ethanol	0.0204	0.0194	0.0194	0.6552	0.0202	0.0281	0.0279
Diesel	0.0332	0.0332	0.0332	10.40	0.0337	0.0338	0.0339
Gasoline	0.0249	0.0239	0.0239	0.5738	0.0261	0.0301	0.0300

Table 5.2: MSFE for $h = 2$.

	Res _{$r=1$}	Res _{$r=2$}	Res _{$r=3$}	Gen _{$r=2$}	RW	VAR	VECM _{$r=2$}
Ethanol	0.0385	0.0366	0.0366	18.65	0.0367	0.0560	0.0556
Diesel	0.0549	0.0550	0.0549	356.81	0.0558	0.0557	0.0560
Gasoline	0.0444	0.0426	0.0426	15.52	0.0457	0.0557	0.0554

Table 5.3: MSFE for $h = 3$.

⁵ In this case, we consider a full-rank version of the Restricted DLM-VECM, which resembles a VAR in levels

	$\text{Res}_{r=1}$	$\text{Res}_{r=2}$	$\text{Res}_{r=3}$	$\text{Gen}_{r=2}$	RW	VAR	$\text{VECM}_{r=2}$
Ethanol	0.0576	0.0554	0.0554	671.43	0.0543	0.0894	0.0886
Diesel	0.0784	0.0785	0.0785	14361.34	0.0796	0.0790	0.0795
Gasoline	0.0655	0.0630	0.0630	560.21	0.0674	0.0849	0.0844

Table 5.4: *MSFE for $h = 4$.*

	$\text{Res}_{r=1}$	$\text{Res}_{r=2}$	$\text{Res}_{r=3}$	$\text{Gen}_{r=2}$	RW	VAR	$\text{VECM}_{r=2}$
Ethanol	0.0770	0.0751	0.0751	27945.39	0.0723	0.1271	0.1259
Diesel	0.1025	0.1029	0.1028	639802.28	0.1046	0.1030	0.1039
Gasoline	0.0890	0.0861	0.0861	23528.79	0.0918	0.1186	0.1177

Table 5.5: *MSFE for $h = 5$.*

	$\text{Res}_{r=1}$	$\text{Res}_{r=2}$	$\text{Res}_{r=3}$	$\text{Gen}_{r=2}$	RW	VAR	$\text{VECM}_{r=2}$
Ethanol	0.0962	0.0944	0.0944	1252886.92	0.0904	0.1685	0.1667
Diesel	0.1292	0.1299	0.1298	29697936.17	0.1316	0.1289	0.1301
Gasoline	0.1147	0.1114	0.1114	1061701.07	0.1184	0.1563	0.1550

Table 5.6: *MSFE for $h = 6$.*

The first question that comes to mind is the excessively high MSFE of the General DLM-VECM. While for $h = 1$ this suggests overfitting, for $h > 1$ we clearly observe an explosive behavior, which indicates nonstationarity. Therefore, a more careful application of the DLM-VECM model should consider the possibility that the eigenvalues of the corresponding VECM regression matrices can be greater than one. For exemplification, Figure 5.5 displays the forecasted values for the Restricted DLM-VECM ($r = 2$), for the General DLM-VECM ($r = 2$) and for the true series at $h = 1$. Notice how the General DLM-VECM values explode during some time periods, resulting in large MSFE.

Additionally, all the Restricted DLM-VECM alternatives outperform the VAR and VECM across all forecasting horizons for all series. With respect to the Random Walk, the results are mixed. In many practical applications, it is known that forecasting prices is difficult and that surpassing the Random Walk benchmark is challenging.

The good performance of the Restricted DLM-VECM suggests that the General DLM-VECM may be of great use if the appropriate restrictions are placed on it. As already outlined, the main difference between them is that some parameters are fixed. And between these two specifications, there is a wide spectrum of possibilities to be explored through restrictions.

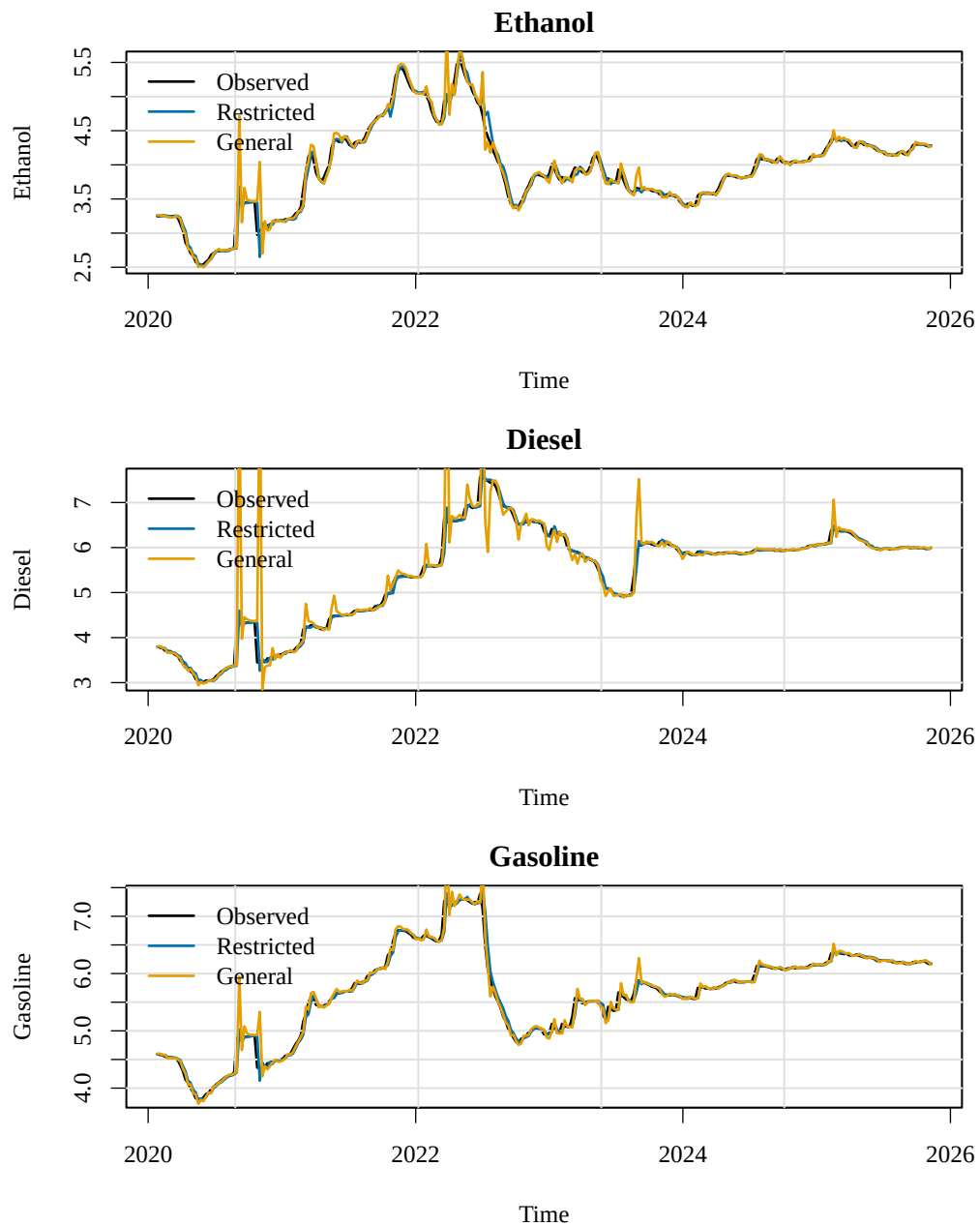


Figure 5.5: *Restricted DLM-VECM vs. General DLM-VECM for $h = 1$.*

Chapter 6

Conclusion

This thesis proposes a new framework for modelling time-varying cointegration through DLMs. In this context, we develop a feasible DLM representation for VECMs that addresses the problems of parameter nonlinearity and the lack of global identification. In this setting, the cointegration space, and not the cointegration vectors, constitutes our main object of interest.

After overcoming the main challenges of the DLM-VECM framework in Chapter 2, we propose two classes of models: the Restricted DLM-VECM and the General DLM-VECM. The Restricted DLM-VECM allows time variation only in the cointegration matrix, thereby treating the remaining VECM parameters as static regression coefficients. The General DLM-VECM, in contrast, treats these regression coefficients as states. With respect to estimation, we derive EM algorithms for all specifications. Moreover, we also consider constrained versions of these EM algorithms that achieve full identification.

To conclude, we apply the DLM-VECM framework to a weekly dataset of prices in the Brazilian retail fuel market. The results indicate that the DLM-VECM approach constitutes an appealing alternative relative to its pairs. Additionally, we have a strong suspicion that incorporating restrictions based on prior knowledge could improve the performance of our models. The Restricted DLM-VECM is only a version of the General DLM-VECM in which a subset of the regression parameters is static, and there are many possibilities in between.

As a continuation of this work, several extensions of the DLM-VECM framework can be considered for future research. For example, one may be interested in studying the impact of imposing restrictions on the parameters and states of the DLM-VECM. Moreover, one can even extend the DMA framework of [KOOP and KOROBILIS, 2013](#) or the Markov-switching approach of [KIM, 1994](#) to accommodate different regimes and cointegration ranks. As exemplified, there are many gaps to be filled in this literature, and we hope that this work has contributed to filling some of them.

Appendix A

Algorithms

Algorithm 1 Basic DLM-VECM: Filtering Algorithm

1: **(0) Inputs:** $r, \{y_t\}_{t=0}^T, b_{0|0}, \Sigma_{0|0}^{(b)}$ and $\Theta_{bas} = \{\rho, H, A\}$.

2: **(1) Prediction and Filtering:**

3: **for** $t = 1$ to T **do**

4: Run the **prediction step**:

$$b_{t|t-1} = \rho b_{t-1|t-1},$$

$$\Sigma_{t|t-1}^{(b)} = \rho^2 \Sigma_{t-1|t-1}^{(b)} + I_{nr}.$$

5: Construct the observation matrix $\tilde{X}_t = (y_{t-1}^\top \otimes A)$.

6: Run the **update step**:

$$b_{t|t} = b_{t|t-1} + \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top (H + \tilde{X}_t \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top)^{-1} (\Delta y_t - \tilde{X}_t b_{t|t-1}),$$

$$\Sigma_{t|t}^{(b)} = \Sigma_{t|t-1}^{(b)} - \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top (H + \tilde{X}_t \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top)^{-1} \tilde{X}_t \Sigma_{t|t-1}^{(b)}.$$

7: **end for**

8: **(2) Outputs:** $\{b_{t|t-1}, \Sigma_{t|t-1}^{(b)}, b_{t|t}, \Sigma_{t|t}^{(b)}\}_{t=0}^T$ and $\{\tilde{X}_t\}_{t=1}^T$.

Algorithm 2 Basic DLM-VECM: Smoothing Algorithm

- 1: **(0) Inputs:** $\{b_{t|t-1}, \Sigma_{t|t-1}^{(b)}, b_{t|t}, \Sigma_{t|t}^{(b)}\}_{t=0}^T, \tilde{X}_T$ and Θ_{bas} .
- 2: **(1) Smoothing:**
- 3: Set the initial conditions $b_{T|T}$ and $\Sigma_{T|T}^{(b)}$
- 4: **for** $t = T - 1$ to 0 **do**
- 5: Run the **smooth step** of the Kalman Filter, using $J_t = \rho \Sigma_{t|t}^{(b)} (\Sigma_{t+1|t}^{(b)})^{-1}$

$$b_{t|T} = b_{t|t} + J_t (b_{t+1|T} - b_{t+1|t}),$$

$$\Sigma_{t|T}^{(b)} = \Sigma_{t|t}^{(b)} + J_t (\Sigma_{t+1|T}^{(b)} - \Sigma_{t+1|t}^{(b)}) J_t^\top.$$

- 6: **end for**
- 7: **(2) Lag-One Covariance Smoother**
- 8: Set the initial condition:

$$\Sigma_{T,T-1|T}^{(b)} = \rho (I - K_T \tilde{X}_T) \Sigma_{T-1|T-1}^{(b)}$$

where $K_T = \Sigma_{T|T-1}^{(b)} \tilde{X}_T^\top (H + \tilde{X}_T \Sigma_{T|T-1}^{(b)} \tilde{X}_T^\top)^{-1}$.

- 9: **for** $t = T$ to 2 **do**

$$\Sigma_{t-1,t-2|T}^{(b)} = \Sigma_{t-1|t-1}^{(b)} J_{t-2}^\top + J_{t-1} \left(\Sigma_{t,t-1|T}^{(b)} - \rho \Sigma_{t-1|t-1}^{(b)} \right) J_{t-2}^\top.$$

- 10: **end for**
 - 11: **(3) Outputs:** $\{b_{t|T}, \Sigma_{t|T}^{(b)}\}_{t=0}^T$ and $\{\Sigma_{t,t-1|T}^{(b)}\}_{t=1}^T$.
-

Algorithm 3 EM Algorithm for the Basic DLM-VECM

- 1: **Initialization:** $r, \Theta_{bas}^{(0)} = \{\rho^{(0)}, H^{(0)}, A^{(0)}\}$.
- 2: **for** $j = 0, 1, 2, \dots$ **until** convergence **do**
- 3: **E-step:**
- 4: Conditional on $\Theta_{bas}^{(j)}$, run the Basic DLM-VECM model through Algorithms 1 and 2 to obtain the filtered and smoothed quantities

$$\mathcal{D}^{(j)} = \left(\{b_{t|T}, \Sigma_{t|T}^{(b)}\}_{t=0}^T; \{\Sigma_{t,t-1|T}^{(b)}\}_{t=1}^T; \{\tilde{X}_t\}_{t=1}^T \right)^{(j)}.$$

- 5: Construct the moment matrices $S_{11}^{(b)}, S_{10}^{(b)}, S_{00}^{(b)}$.
- 6: **M-step:** Update the parameters in $\Theta_{bas}^{(j)}$ to obtain $\Theta_{bas}^{(j+1)}$:

$$\rho^{(j+1)} = \frac{\text{tr}(S_{10}^{(b)})}{\text{tr}(S_{00}^{(b)})},$$

$$H^{(j+1)} = \frac{1}{T} \sum_{t=1}^T \left(\Delta y_t \Delta y_t^\top - 2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top + \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top \right),$$

$$A^{(j+1)} = \left(\sum_{t=1}^T \Delta y_t b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top \right) \times \left(\sum_{t=1}^T (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top \right)^{-1}.$$

- 7: **Convergence:** Compute the observed Likelihood $L_Y(\Theta_{bas}^{(j+1)})$ to verify if convergence is attained.
 - 8: **end for**
-

Algorithm 4 Restricted DLM-VECM: Filtering Algorithm

- 1: **(0) Inputs:** $r, p, \{z_t\}_{t=1}^T, \{y_t\}_{t=0}^T, b_{0|0}, \Sigma_{0|0}^{(b)}$ and $\Theta_{res} = \{\rho, H, A, \Psi\}$.
 - 2: **(1) Prediction and Filtering:**
 - 3: **for** $t = 1$ to T **do**
 - 4: Run the **prediction step**:

$$b_{t|t-1} = \rho b_{t-1|t-1},$$

$$\Sigma_{t|t-1}^{(b)} = \rho^2 \Sigma_{t-1|t-1}^{(b)} + I_{nr}.$$
 - 5: Construct the observation matrix $\tilde{X}_t = (y_t^\top \otimes A)$.
 - 6: Construct the exogenous matrix Z_t .
 - 7: Run the **update step**:

$$b_{t|t} = b_{t|t-1} + \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top (H + \tilde{X}_t \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top)^{-1} (\Delta y_t - Z_t \Psi - \tilde{X}_t b_{t|t-1}),$$

$$\Sigma_{t|t}^{(b)} = \Sigma_{t|t-1}^{(b)} - \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top (H + \tilde{X}_t \Sigma_{t|t-1}^{(b)} \tilde{X}_t^\top)^{-1} \tilde{X}_t \Sigma_{t|t-1}^{(b)}.$$
 - 8: **end for**
 - 9: **(2) Outputs:** $\{b_{t|t-1}, \Sigma_{t|t-1}^{(b)}, b_{t|t}, \Sigma_{t|t}^{(b)}\}_{t=0}^T, \{Z_t\}_{t=1}^T$ and $\{\tilde{X}_t\}_{t=1}^T$.
-

Algorithm 5 Restricted DLM-VECM: Smoothing Algorithm

- 1: **(0) Inputs:** $\{b_{t|t-1}, \Sigma_{t|t-1}^{(b)}, b_{t|t}, \Sigma_{t|t}^{(b)}\}_{t=0}^T, \tilde{X}_T$ and Θ_{res} .
- 2: **(1) Smoothing:**
- 3: Set the initial conditions $b_{T|T}$ and $\Sigma_{T|T}^{(b)}$
- 4: **for** $t = T - 1$ to 0 **do**
- 5: Run the **smooth step** of the Kalman Filter, using $J_t = \rho \Sigma_{t|t}^{(b)} (\Sigma_{t+1|t}^{(b)})^{-1}$

$$b_{t|T} = b_{t|t} + J_t (b_{t+1|T} - b_{t+1|t}),$$

$$\Sigma_{t|T}^{(b)} = \Sigma_{t|t}^{(b)} + J_t (\Sigma_{t+1|T}^{(b)} - \Sigma_{t+1|t}^{(b)}) J_t^\top.$$

- 6: **end for**
- 7: **(2) Lag-One Covariance Smoother**
- 8: Set the initial condition:

$$\Sigma_{T, T-1|T}^{(b)} = \rho (I - K_T \tilde{X}_T) \Sigma_{T-1|T-1}^{(b)}$$

$$\text{where } K_T = \Sigma_{T|T-1}^{(b)} \tilde{X}_T^\top (H + \tilde{X}_T \Sigma_{T|T-1}^{(b)} \tilde{X}_T^\top)^{-1}.$$

- 9: **for** $t = T$ to 2 **do**

$$\Sigma_{t-1, t-2|T}^{(b)} = \Sigma_{t-1|t-1}^{(b)} J_{t-2}^\top + J_{t-1} \left(\Sigma_{t, t-1|T}^{(b)} - \rho \Sigma_{t-1|t-1}^{(b)} \right) J_{t-2}^\top.$$

- 10: **end for**

- 11: **(3) Outputs:** $\{b_{t|T}, \Sigma_{t|T}^{(b)}\}_{t=0}^T$ and $\{\Sigma_{t, t-1|T}^{(b)}\}_{t=1}^T$.
-

Algorithm 6 EM Algorithm for the Restricted DLM-VECM

- 1: **Initialization:** $r, p, \Theta_{res}^{(0)} = \{\rho^{(0)}, H^{(0)}, A^{(0)}, \Psi^{(0)}\}$.
- 2: **for** $j = 0, 1, 2, \dots$ **until convergence do**
- 3: **E-step:**
- 4: Conditional on $\Theta_{res}^{(j)}$, run the Restricted DLM-VECM model through Algorithms 4 and 5 to obtain the filtered and smoothed quantities

$$\mathcal{D}^{(j)} = \left(\{b_{t|T}, \Sigma_{t|T}^{(b)}\}_{t=0}^T; \{\Sigma_{t,t-1|T}^{(b)}\}_{t=1}^T; \{Z_t\}_{t=1}^T; \{\tilde{X}_t\}_{t=1}^T \right)^{(j)}.$$

- 5: Construct the moment matrices $S_{11}^{(b)}, S_{10}^{(b)}, S_{00}^{(b)}$.
- 6: **M-step:** Update the parameters in $\Theta_{res}^{(j)}$ to obtain $\Theta_{res}^{(j+1)}$:

$$\rho^{(j+1)} = \frac{\text{tr}(S_{10}^{(b)})}{\text{tr}(S_{00}^{(b)})},$$

$$H^{(j+1)} = \frac{1}{T} \sum_{t=1}^T \left(\Delta y_t \Delta y_t^\top - 2 \Delta y_t b_{t|T}^\top \tilde{X}_t^\top - 2 \Delta y_t \Psi^{(j)\top} Z_t^\top + 2 \tilde{X}_t b_{t|T} \Psi^{(j)\top} Z_t^\top + \tilde{X}_t (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) \tilde{X}_t^\top + Z_t \Psi^{(j)} \Psi^{(j)\top} Z_t^\top \right),$$

$$\Psi^{(j+1)} = \left(\sum_{t=1}^T Z_t^\top (H^{(j+1)})^{-1} Z_t \right)^{-1} \left(\sum_{t=1}^T Z_t^\top (H^{(j+1)})^{-1} (\Delta y_t - \tilde{X}_t b_{t|T}) \right),$$

$$A^{(j+1)} = \left(\sum_{t=1}^T (\Delta y_t - Z_t \Psi^{(j+1)}) b_{t|T}^\top (y_{t-1}^\top \otimes I_r)^\top \right) \times \left(\sum_{t=1}^T (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top \right)^{-1}.$$

- 7: **Convergence:** Compute the observed Likelihood $L_Y(\Theta_{res}^{(j+1)})$ to verify if convergence is attained.
 - 8: **end for**
-

Algorithm 7 General DLM-VECM: Filtering Algorithm

1: **(0) Inputs:** $r, p, \{z_t\}_{t=1}^T, \{y_t\}_{t=0}^T, \theta_{0|0}, \Sigma_{0|0}$ and $\Theta_{gen} = \{\Psi_{0|0}, \Sigma_{0|0}^{(\Psi)}, \rho, H, A, Q^{(\Psi)}\}$.

2: **(1) Prediction and Filtering:**

3: **for** $t = 1$ to T **do**

4: Run the **prediction step**:

$$\begin{aligned}\theta_{t|t-1} &= G\theta_{t-1|t-1}, \\ \Sigma_{t|t-1} &= G\Sigma_{t-1|t-1}G^\top + Q,\end{aligned}$$

5: Construct matrices $\tilde{X}_t = (y_{t-1}^\top \otimes A)$ and Z_t .

6: Build the measurement matrix $F_t = (\tilde{X}_t \ Z_t)$.

7: Run the **update step**:

$$\begin{aligned}\theta_{t|t} &= \theta_{t|t-1} + \Sigma_{t|t-1}F_t^\top (H + F_t\Sigma_{t|t-1}F_t^\top)^{-1} (\Delta y_t - F_t\theta_{t|t-1}), \\ \Sigma_{t|t} &= \Sigma_{t|t-1} - \Sigma_{t|t-1}F_t^\top (H + F_t\Sigma_{t|t-1}F_t^\top)^{-1} F_t\Sigma_{t|t-1}.\end{aligned}$$

8: **end for**

9: **(2) Outputs:** $\{\theta_{t|t-1}, \Sigma_{t|t-1}, \theta_{t|t}, \Sigma_{t|t}\}_{t=0}^T$ and $\{F_t\}_{t=1}^T$.

Algorithm 8 General DLM-VECM: Smoothing Algorithm

1: **(0) Inputs:** $\{\theta_{t|t-1}, \Sigma_{t|t-1}, \theta_{t|t}, \Sigma_{t|t}\}_{t=0}^T$ and Θ_{gen} .

2: **(1) Smoothing:**

3: Set the initial conditions $\theta_{T|T}$, F_T and $\Sigma_{T|T}$.

4: **for** $t = T - 1$ to 0 **do**

5: Compute the smoothing gain

$$J_t = \Sigma_{t|t} G^\top (\Sigma_{t+1|t})^{-1}.$$

6: Run the **smooth step**:

$$\begin{aligned}\theta_{t|T} &= \theta_{t|t} + J_t (\theta_{t+1|T} - \theta_{t+1|t}), \\ \Sigma_{t|T} &= \Sigma_{t|t} + J_t (\Sigma_{t+1|T} - \Sigma_{t+1|t}) J_t^\top.\end{aligned}$$

7: **end for**

8: **(2) Lag-One Covariance Smoother**

9: Set the initial condition:

$$\Sigma_{T,T-1|T} = (I - K_T F_T) G \Sigma_{T-1|T-1},$$

where

$$K_T = \Sigma_{T|T-1} F_T^\top (H + F_T \Sigma_{T|T-1} F_T^\top)^{-1}.$$

10: **for** $t = T$ to 2 **do**

$$\Sigma_{t-1,t-2|T} = \Sigma_{t-1|t-1} J_{t-2}^\top + J_{t-1} (\Sigma_{t,t-1|T} - G \Sigma_{t-1|t-1}) J_{t-2}^\top.$$

11: **end for**

12: **(3) Outputs:** $\{\theta_{t|T}, \Sigma_{t|T}\}_{t=0}^T$ and $\{\Sigma_{t,t-1|T}\}_{t=1}^T$.

Algorithm 9 EM Algorithm for the General DLM-VECM

- 1: **Initialization:** $r, p, \Theta_{gen}^{(0)} = \{\Psi_{0|0}^{(0)}, \Sigma_{0|0}^{(\Psi)(0)}, \rho^{(0)}, H^{(0)}, A^{(0)}, Q^{(\Psi)(0)}\}$.
- 2: **for** $j = 0, 1, 2, \dots$ **until** convergence **do**
- 3: **E-step:**
- 4: Conditional on $\Theta_{gen}^{(j)}$, run the General DLM-VECM model through Algorithms 7 and 8 to obtain the filtered and smoothed quantities

$$\mathcal{D}^{(j)} = \left(\{\theta_{t|T}, \Sigma_{t|T}^{(\theta)}\}_{t=0}^T; \{\Sigma_{t,t-1|T}^{(\theta)}\}_{t=1}^T; \{Z_t\}_{t=1}^T; \{\tilde{X}_t\}_{t=1}^T; \{F_t\}_{t=1}^T \right)^{(j)}.$$

- 5: Construct the moment matrices $S_{11}^{(\theta)}, S_{10}^{(\theta)}, S_{00}^{(\theta)}$ and its corresponding block components associated with b and Ψ . Also extract the similar block components of b and Ψ from $\{\Sigma_{t|T}^{(\theta)}\}_{t=0}^T$ and $\{\Sigma_{t,t-1|T}^{(\theta)}\}_{t=1}^T$.
- 6: **M-step:** Update the parameters in $\Theta_{gen}^{(j)}$ to obtain $\Theta_{gen}^{(j+1)}$:

$$\begin{aligned} \Psi_{0|0}^{(j+1)} &= \begin{pmatrix} 0_{k \times nr} & I_k \end{pmatrix} \theta_{0|T}, \\ \Sigma_{0|0}^{(\Psi)(j+1)} &= \begin{pmatrix} 0_{k \times nr} & I_k \end{pmatrix} \Sigma_{0|T}^{(\theta)} \begin{pmatrix} 0_{nr \times k} \\ I_k \end{pmatrix}, \\ \rho^{(j+1)} &= \frac{\text{tr}(S_{10}^{(b)})}{\text{tr}(S_{00}^{(b)})}, \\ Q^{(\Psi)(j+1)} &= \frac{1}{T} \left(S_{11}^{(\Psi)} - S_{10}^{(\Psi)} - S_{10}^{(\Psi)\top} + S_{00}^{(\Psi)} \right), \\ H^{(j+1)} &= \frac{1}{T} \sum_{t=1}^T \left(\Delta y_t \Delta y_t^\top - 2 \Delta y_t \theta_{t|T}^\top F_t^\top + F_t (\theta_{t|T} \theta_{t|T}^\top + \Sigma_{t|T}^{(\theta)}) F_t^\top \right), \\ A^{(j+1)} &= \left(\sum_{t=1}^T (\Delta y_t b_{t|T}^\top - Z_t \Psi_{t|T} b_{t|T}^\top - Z_t \Sigma_{t|T}^{(\Psi,b)}) (y_{t-1}^\top \otimes I_r)^\top \right) \\ &\quad \times \left(\sum_{t=1}^T (y_{t-1}^\top \otimes I_r) (b_{t|T} b_{t|T}^\top + \Sigma_{t|T}^{(b)}) (y_{t-1}^\top \otimes I_r)^\top \right)^{-1}. \end{aligned}$$

- 7: **Convergence:** Compute the observed Likelihood $L_Y(\Theta_{gen}^{(j+1)})$ to verify if convergence is attained.
 - 8: **end for**
-

Appendix B

Kalman Filter and Smoother

Both the Kalman Filter and the Kalman Smoother are complementary algorithms for inference on the unobserved states of a linear-Gaussian state space model. They are recursive procedures that provide minimum mean-squared error (MSE) estimates of the latent states, conditional on the available information set $\mathcal{F}_s$.

We start with a general state space model formulation:

$$y_t = M_t \theta_t + Z_t u_t + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, H), \quad (\text{B.1})$$

$$\theta_{t+1} = \Phi \theta_t + P \eta_t, \quad \eta_t \stackrel{i.i.d.}{\sim} N(0, Q), \quad (\text{B.2})$$

where the elements are defined as:

- y_t is the $n \times 1$ observation vector.
- M_t is the $n \times k$ observation/measurement matrix.
- θ_t is the $m \times 1$ state vector.
- Z_t is the $n \times q$ exogenous matrix.
- u_t is the $q \times 1$ exogenous vector.
- ε_t is the $n \times 1$ observation error.
- Φ is the $m \times m$ state transition matrix.
- η_t is the $k \times 1$ state disturbance.
- P is an $m \times k$ loading matrix for the state disturbance. It allows the state disturbances to have a smaller dimension than the state vector ($k \leq m$). It is specified when some subset of states do not evolve over time according to the state equation. When all states are time-varying, it is usually specified as $P = I_m$.

Given an agnostic information set $\mathcal{F}_s$, we denote the state estimates by

$$\theta_{t|s} = E[\theta_t \mid \mathcal{F}_s],$$

with corresponding covariance matrix

$$\Sigma_{t|s} = E[(\theta_t - \theta_{t|s})(\theta_t - \theta_{t|s})^\top | \mathcal{F}_s],$$

and cross-covariance matrices of lag τ defined as

$$\Sigma_{t, t-\tau|s} = E[(\theta_t - \theta_{t|s})(\theta_{t-\tau} - \theta_{t-\tau|s})^\top | \mathcal{F}_s].$$

While the Kalman Filter is used for online estimation, thus conditioning on $\mathcal{F}_t$, the Kalman Smoother conditions on the whole sample, which is represented by $\mathcal{F}_T$.

In the following developments, we present the related Kalman filtering and smoothing recursions for the DLM specified in equations (B.1) and (B.2). For the reader interested in the full analytical derivations of the Kalman Filter and Kalman Smoother, we recommend the books by SHUMWAY and STOFFER, 2006 and HAMILTON, 1994. Additionally, we refer to PETRIS *et al.*, 2009 for a Bayesian perspective on Dynamic Linear Models.

Filtering

The **Kalman Filter** procedure for the DLM in equations (B.1) and (B.2) depends on the following distributions:

$$\theta_t | \mathcal{F}_{t-1} \sim \mathcal{N}(\theta_{t|t-1}, \Sigma_{t|t-1}) \quad (\text{B.3})$$

$$\theta_t | \mathcal{F}_t \sim \mathcal{N}(\theta_{t|t}, \Sigma_{t|t}) \quad (\text{B.4})$$

$$y_t | \mathcal{F}_{t-1} \sim \mathcal{N}(M_t \theta_{t|t-1} + Z_t u_t, H + M_t \Sigma_{t|t-1} M_t^\top) \quad (\text{B.5})$$

By conditioning on the DLM parameters Q and H , and on a prior distribution of the states

$$\theta_0 \sim \mathcal{N}(\theta_{0|0}, \Sigma_{0|0}), \quad (\text{B.6})$$

the one-step-ahead predicted and filtered states $\theta_{t|t-1}$, $\theta_{t|t}$, along with their associated covariance matrices $\Sigma_{t|t}$ and $\Sigma_{t|t-1}$, are recursively computed through the following equations, for $t = 1, \dots, T$:

$$\theta_{t|t-1} = \Phi \theta_{t-1|t-1} \quad (\text{B.7})$$

$$\Sigma_{t|t-1} = \Phi \Sigma_{t-1|t-1} \Phi^\top + P Q P^\top, \quad (\text{B.8})$$

$$\theta_{t|t} = \theta_{t|t-1} + \Sigma_{t|t-1} M_t^\top (H + M_t \Sigma_{t|t-1} M_t^\top)^{-1} (y_t - M_t \theta_{t|t-1} - Z_t u_t), \quad (\text{B.9})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1} M_t^\top (H + M_t \Sigma_{t|t-1} M_t^\top)^{-1} M_t \Sigma_{t|t-1}. \quad (\text{B.10})$$

where the quantity

$$K_t = \Sigma_{t|t-1} M_t^\top (H + M_t \Sigma_{t|t-1} M_t^\top)^{-1} \quad (\text{B.11})$$

is called the *Kalman gain*.

Estimation of Q and H is usually performed by the Expectation Maximization method,

or by a numerical Newton-Raphson procedure. Regarding the prior distribution in (B.6), it is usually set as a non-informative prior with mean zero and large variance, or even as the Minnesota prior [LITTERMAN, 1986, BAÑBURA, GIANNONE, *et al.*, 2010]. Additionally, an implementation of this algorithm may require a computationally stable alternative, such as the SVD Kalman Filter of WANG *et al.*, 1992.

Smoothing

Given the filtered states and covariances obtained from equations (B.7)–(B.10) for $t = 1, \dots, T$, the **Kalman Smoother** with initial conditions $\theta_{T|T}$ and $\Sigma_{T|T}$ is defined by the following recursions, for $t = T - 1, T - 2, \dots, 0$:

$$\theta_{t|T} = \theta_{t|t} + J_t (\theta_{t+1|T} - \theta_{t+1|t}), \quad (\text{B.12})$$

$$\Sigma_{t|T} = \Sigma_{t|t} + J_t (\Sigma_{t+1|T} - \Sigma_{t+1|t}) J_t^\top, \quad (\text{B.13})$$

where

$$J_t = \Sigma_{t|t} \Phi^\top (\Sigma_{t+1|t})^{-1} \quad (\text{B.14})$$

is called the *smoothing gain*.

Additionally, we will need the **lag-one covariance smoother** matrices for the EM procedure. With initial condition

$$\Sigma_{T,T-1|T} = (I - K_T M_T) \Phi \Sigma_{T-1|T-1}, \quad (\text{B.15})$$

the lag-one covariance smoother matrices for $t = T, T - 1, \dots, 2$ are given by

$$\Sigma_{t-1,t-2|T} = \Sigma_{t-1|t-1} J_{t-2}^\top + J_{t-1} \left(\Sigma_{t,t-1|T} - \Phi \Sigma_{t-1|t-1} \right) J_{t-2}^\top. \quad (\text{B.16})$$

This completes the set of recursions needed for estimating the smoothed states.

Appendix C

Supporting Results from KOOP, LEON-GONZALEZ, *et al.*, 2011

This appendix section summarizes the main results of KOOP, LEON-GONZALEZ, *et al.*, 2011 that are useful for our derivations. In particular, KOOP, LEON-GONZALEZ, *et al.* show that modelling the unrestricted cointegration matrix B_t through the restricted state equation in (2.5), along with an initial condition as in (2.6), results in an evolution of the cointegration space $\mathbf{p}_t$ that displays three desirable properties.

The first property is that the conditional cointegration space, denoted by $\mathbf{p}_t \mid \mathbf{p}_{t-1}$, has a distribution that is centered on $\mathbf{p}_{t-1}$. By “centered,” they mean it in a modal and in an expectation sense. The second property is that the change in location from $\mathbf{p}_{t-1}$ to $\mathbf{p}_t$ is gradual, being ruled by the concentration parameter $\rho^2 \kappa_t^2 = \rho^2(B_t^\top B_t)$. The third property states that prior beliefs about the marginal distribution of $\mathbf{p}_t$ can be incorporated into the model. In this context, KOOP, LEON-GONZALEZ, *et al.* leverage STRACHAN and INDER, 2004 to characterize the uninformative case. In this case, $b_t \sim \mathcal{N}(0, cI_{nr})$ implies a marginal uniform distribution for β_t on the Stiefel manifold and a uniform distribution for $\mathbf{p}_t$ on the Grassmann manifold, for any $c > 0$.¹

The first property is supported by Propositions 1 and 2, while the second property is supported by Propositions 3 and 4 in the referred article. The use of the third property in this paper is limited to the uninformative case and is not developed further. For a deeper understanding, we recommend checking Proposition 5 in the working paper of KOOP, LEON-GONZALEZ, *et al.*, 2011, along with STRACHAN and INDER, 2004.

To derive their Propositions, KOOP, LEON-GONZALEZ, *et al.* choose the semi-orthogonality restriction² to identify the cointegration space. They denote this semi-

¹ The Stiefel manifold is represented by the set of $n \times r$ orthonormal matrices, that is $V_r(\mathbb{R}^n) = \{\beta \in \mathbb{R}^{n \times r} : \beta^\top \beta = I_r\}$, while the Grassmann manifold corresponds to the set of r -dimensional subspaces of $\mathbb{R}^n$, being denoted by $G(r, n) = \{S \subset \mathbb{R}^n : \dim(S) = r\}$.

² Read their working paper to understand why this restriction does not limit the estimable cointegration space and is therefore preferable to its alternatives.

orthogonal matrix by β_t , which is obtainable from following transformation:

$$\beta_t = B_t(\kappa_t)^{-1}, \quad (\text{C.1})$$

$$\kappa_t = (B_t^\top B_t)^{1/2}, \quad (\text{C.2})$$

implying that $\beta_t^\top \beta_t = I_r$. It is important to notice that β_t is not unique. Indeed, for any orthonormal matrix Q , $\beta_t Q$ spans the same column space (cointegration space) as β_t . In this context, an unique representation for the cointegration space $\mathbf{p}_t$ is obtainable from the matrix product

$$P_t = \beta_t Q Q^\top \beta_t^\top = \beta_t \beta_t^\top, \quad (\text{C.3})$$

where P_t is an orthogonal idempotent matrix of rank r . This matrix uniquely represents the Grassman manifold associated with the cointegration space $\mathbf{p}_t$. From these preliminaries, KOOP, LEON-GONZALEZ, *et al.* aim to find the conditional distribution for P_t , which is equivalent to finding the distribution for $\mathbf{p}_t \mid \mathbf{p}_{t-1}$.

They start from equation (2.5), from which the conditional distribution of $b_t \mid b_{t-1}$ is given by

$$b_t \mid b_{t-1} \sim \mathcal{N}(\rho b_{t-1}, I_{nr}). \quad (\text{C.4})$$

This implies that

$$B_t \mid B_{t-1} \sim \mathcal{N}_{n,r}(\rho B_{t-1}, I_n, I_r), \quad (\text{C.5})$$

which corresponds to a matrix normal distribution. Proceeding, KOOP, LEON-GONZALEZ, *et al.* translate the distribution in (C.5) to the corresponding orthonormal basis for the column space of B_t ,³ which is uniquely represented by P_t . For this purpose, they apply Theorem 2.4.9 from CHIKUSE, 2003, which states that the conditional distribution for P_t (or equivalently $\mathbf{p}_t \mid \mathbf{p}_{t-1}$) follows the orthogonal projective Gaussian distribution with parameter

$$F_t = \rho B_{t-1} (I_r)^{-1} B_{t-1}^\top \rho^\top = \beta_{t-1} \rho^2 \kappa_{t-1}^2 \beta_{t-1}^\top. \quad (\text{C.6})$$

This corresponds to the following density

$$f(P_t \mid F_t) = \exp\left(-\frac{1}{2} \text{tr}(F_t)\right) {}_1F_1\left(\frac{n}{2}; \frac{r}{2}; \frac{1}{2} F_t P_t\right), \quad (\text{C.7})$$

where ${}_pF_q$ denotes a hypergeometric function of matrix argument.

Having derived the distribution for the cointegration space, KOOP, LEON-GONZALEZ, *et al.* state four propositions that ensure the desirable properties for the conditional cointegration space $\mathbf{p}_t \mid \mathbf{p}_{t-1}$. All are proved in the Appendix of the referenced paper.

Proposition 1. *Since $\mathbf{p}_t \mid \mathbf{p}_{t-1} = \text{sp}(\beta_t \mid \beta_{t-1})$ follows an $\text{OPG}(F_t)$ distribution with parameter $F_t = \beta_{t-1} \rho^2 \kappa_{t-1}^2 \beta_{t-1}^\top$, the density function of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ is maximized at $\text{sp}(\beta_{t-1})$.*

³ The column space of B_t is the cointegration space.

This first proposition argues that $\mathbf{p}_t$ is centered over $\mathbf{p}_{t-1}$ in a modal sense. Additionally, this result highlights the necessity of an isotropic covariance matrix in the state equation. If, for instance, $B_t \mid B_{t-1} \sim \mathcal{N}_{n,r}(\rho B_{t-1}, I_n, \Sigma)$ with $\Sigma^{-1} = LL^\top$, then $\mathbf{p}_t$ would have an orthogonal projective Gaussian distribution with parameter

$$\begin{aligned} F_t &= \rho^2 B_{t-1} \Sigma^{-1} B_{t-1}^\top \\ &= \rho^2 B_{t-1} L L^\top B_{t-1}^\top \\ &= \rho^2 \tilde{B}_{t-1} \tilde{B}_{t-1}^\top, \end{aligned}$$

where $\tilde{B}_{t-1} = B_{t-1} L$. This also corresponds to the following matrix-normal distribution for $\tilde{B}_t \mid \tilde{B}_{t-1}$:

$$\tilde{B}_t \mid \tilde{B}_{t-1} \sim \mathcal{N}_{n,r}(\rho \tilde{B}_{t-1}, I_n, I_r). \quad (\text{C.8})$$

Consequently, this induced orthogonal projective Gaussian distribution for $\mathbf{p}_t = \text{sp}(\tilde{B}_t)$ has its mode at $\text{sp}(\tilde{B}_{t-1})$. However since $\text{sp}(\tilde{B}_{t-1}) = \text{sp}(B_{t-1} L) \neq \text{sp}(B_{t-1})$ unless $LL^\top = I_r$, the conditional distribution of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ is not mode centered at $\mathbf{p}_{t-1}$ when Σ is not the identity.

Proposition 2. *Since $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ follows an $\text{OPG}(F_t)$ distribution with parameter $F_t = \beta_{t-1} \rho^2 \kappa_{t-1}^2 \beta_{t-1}^\top$, it follows that $E(\mathbf{p}_t \mid \mathbf{p}_{t-1}) = \text{sp}(\beta_{t-1})$.*

This second proposition argues that $\mathbf{p}_t$ is centered over $\mathbf{p}_{t-1}$ in an expectation sense. To this end, they use the location measure proposed by VILLANI, 2006, based on the Frobenius norm.

Proposition 3. *Assume $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ follows an $\text{OPG}(F_t)$ distribution with parameter $F_t = \beta_{t-1} \rho^2 \kappa_{t-1}^2 \beta_{t-1}^\top$. Then:*

1. *The value of the density function of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ at the mode depends only on the eigenvalues of $K_t = \rho^2 \kappa_{t-1}^2$.*
2. *The value of the density function of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ at the mode tends to infinity if any of the eigenvalues of K_t tends to infinity.*

This third proposition states that $K_t = \rho^2 \kappa_{t-1}^2$ controls the density of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ around $\mathbf{p}_{t-1}$. Thus, K_t acts as a time-varying concentration parameter for the conditional cointegration space.

Proposition 4. *Suppose $\{\beta_t^* : t = 1, \dots, T\}$ follows the process described by (2.5) and (2.6), with $|\rho| < 1$. Then:*

1. *The marginal distribution for K_t is a Wishart distribution of dimension r with n degrees of freedom and scale matrix $I_r \frac{\rho^2}{1-\rho^2}$.*
2. *The unconditional expectation for K_t is given by $E(K_t) = I_r \frac{n\rho^2}{1-\rho^2}$.*
3. *The conditional expectation for K_t is given by $E(K_t \mid K_{t-1}, \dots, K_2) = \rho^2 K_{t-1} + (1 - \rho^2)E(K_t)$.*
4. *The correlation between the (i, j) element of K_t and the (k, l) element of K_{t-h} is zero unless $i = k$ and $j = l$.*

5. *The correlation between the (i, j) element of K_t and the (i, j) element of K_{t-h} is ρ^{2h} . That is, it decays geometrically*

This fourth proposition summarizes the properties for the joint prior for the concentration parameters $(K_2, \dots, K_T)$. Our main interest lies in item 2, which states that the expected concentration of $\mathbf{p}_t \mid \mathbf{p}_{t-1}$ around $\mathbf{p}_{t-1}$ is time-invariant. Additionally, it shows that when $\rho = 1$, the expected value of the concentration parameter diverges to infinity.

Appendix D

Understanding ρ Through Simulation

This appendix section aims to understand the magnitude of the dispersion in the DLM-VECM implied by ρ . For this purpose, we start by simulating the state equation (2.5) with $n = 2$ and $r = 1$ over $T = 1000$ observations for the values $\rho \in \{0.99, 0.999, 0.9999, 0.99999\}$. Figure D.1 displays the normalized coefficients, that is, the ratio $b_{2,t}/b_{1,t}$ for these simulations.

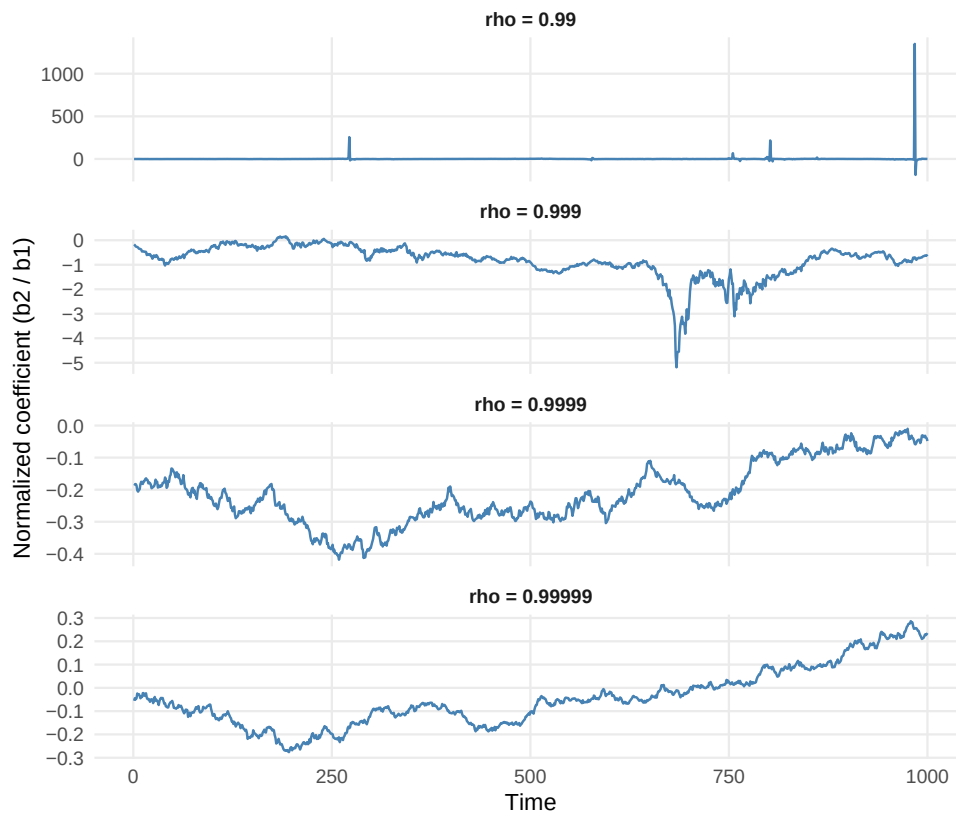


Figure D.1: Normalized paths of $b_{2,t}/b_{1,t}$ for different ρ 's.

Notice that values of ρ closer to one imply a smaller degree of dispersion in the cointegration space. Additionally, observe that the value $\rho = 0.99$ implies a substantial amount of dispersion, which may lead to spikes in its distribution caused by values of $b_{1,t}$ that are too close to zero.

From these simulated quantities, we may obtain the angles implied by each cointegration vector. When $n = 2$, this angle corresponds to the arctangent of the straight line through the origin with slope $-b_{2,t}/b_{1,t}$, that is:

$$\theta_t = \arctan\left(-\frac{b_{2,t}}{b_{1,t}}\right). \quad (\text{D.1})$$

Figure D.2 displays the time evolution of θ_t for different values of ρ .

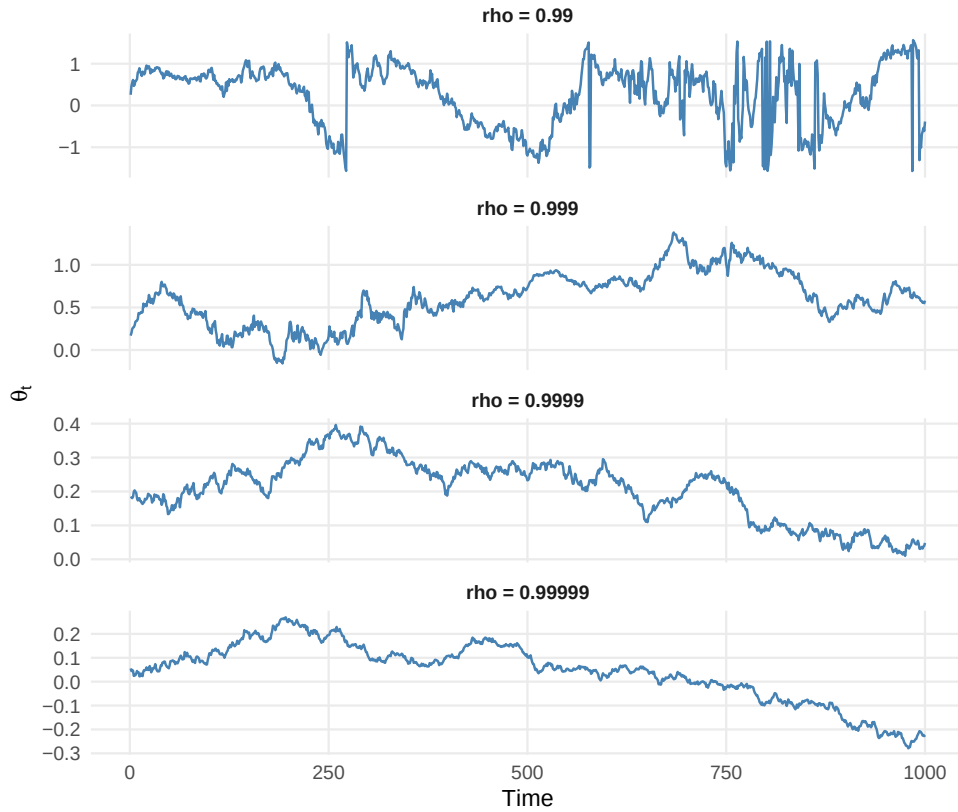


Figure D.2: Paths of θ_t for different ρ 's.

A measure of dispersion may then be given by the difference of these angles. Figure D.3 below plots a kernel density estimator (KDE) of these angle differences $\Delta\theta_t$ for different values of ρ .

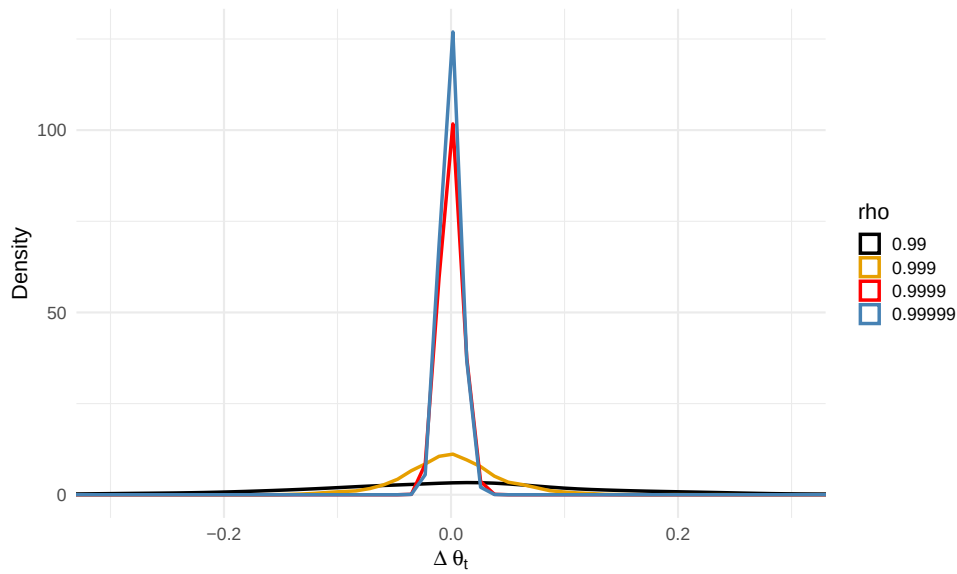


Figure D.3: KDE's of $\Delta\theta_t$ for different ρ 's.

The reader may observe that 0.99 implies an unreasonable amount of variation for certain economic systems. For example, an angle change of 0.2 radians corresponds to approximately $0.2 \times \frac{180}{\pi} \approx 11.5^\circ$. For the weekly series used in Chapter 5, this implies an implausible rotation of the cointegration direction between periods. This is why we put a lower bound of 0.999 on the EM algorithms used in this paper, which may help avoid unwanted optimization behavior. Additionally, we set the upper bound of ρ to 0.999999 to avoid values of ρ greater or equal to one.

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